Never-Realized Capital Gains

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In realization-based tax systems, capital gains are not taxed until the asset is sold. Policies of "stepped-up basis" make capital gains deferral permanent when appreciated assets pass in inheritance. We construct novel measures of unrealized and inherited capital gains in Norway. Unrealized gains are large, top-heavy, and poorly proxied by realized gains. Gains passing through inheritance equal 60% of the realized gains tax base. Two difference-in-differences designs show Norway's 2006 removal of stepped-up basis raised capital gains tax revenue by more than 17%. Yet realizations are small relative to accrual, so the stock of unrealized gains continues to grow after the reform.

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Introduction

Over the past decade, numerous proposals have sought to expand or reform the taxation of wealth and capital. These proposals are often motivated by the concern that income tax systems leave many capital gains permanently untaxed. In the US, UK, and ten other OECD countries, this concern is exemplified by "stepped-up basis" policies, in which capital gains left in inheritance (rather than realized in a sale by their original owner) are exempted from the income tax base.[1] Debate on these issues is divided over whether unrealized capital gains constitute a meaningful gap in the tax base, and, if so, whether the problem is stepped-up basis specifically or features of realization-based income taxation more generally. Several primitives are central to this debate and largely previously unmeasured: the magnitude of unrealized and never-realized capital gains, the distribution of such gains across assets and owners, and the extent to which stepped-up basis causally inhibits realization.

This paper uses Norway as an empirical setting to combine high-quality administrative data, a novel approach to measuring unrealized gains, and a rare policy experiment that removed stepped-up basis. We address two sets of questions. First, descriptively: how large is the stock of unrealized capital gains, how much passes across generations without realization by the original owner, who holds these gains, and to what types of assets are they attributable? Second, causally: how does removing stepped-up basis affect realizations and intergenerational transfers, and how large are these responses relative to the accumulated stock of unrealized capital gains?

Two key challenges have inhibited prior empirical study of these questions. The first challenge is that, globally, tax records often include information on realized capital gains, but generally do not measure unrealized gains or the gains embedded in gifts and bequests. As a result, little is known about which gains remain outside the realized tax base, which gains may eventually be inherited, or who benefits from stepped-up basis. We overcome this challenge by reconstructing detailed capital gains records for assets held by Norwegian households. We link Norwegian administrative data on household wealth and income, asset

[1] Chile, France, Korea, Lithuania, Portugal, Slovenia, Spain, the United Kingdom, and the United States all apply step-up to most deathtime inheritance transfers; while Denmark, Finland, and Hungary have more complicated regimes in which step-up applies to only some types of inherited assets (OECD, 2021). Most countries do not apply step-up to capital gains passed as *inter-vivos* gifts.

purchases and sales, and current asset values at the individual-asset-year level. Although Norwegian tax records only measure realized capital gains (RCG), our linked data allow us to construct measures of unrealized capital gains (UCG) and inherited capital gains (ICG) in listed stock, private businesses, and real estate holdings.

The second challenge is policy variation. Despite many proposals to modify the tax treatment of inherited capital gains, few policy changes have actually been enacted in a way that allows for empirical evaluation of the resulting behavioral effects.[2] In this paper, we overcome this challenge by studying Norway, the only country we are aware of that has permanently and substantially modified its treatment of inherited capital gains in the last four decades by removing stepped-up basis.[3]

In the first part of the paper, we establish four facts about unrealized and inherited capital gains in Norway prior to the removal of stepped-up basis. First, unrealized capital gains are large and important for understanding top wealth: 12–14% of total household wealth is composed of unrealized capital gains in assets subject to the capital gains tax, and that fraction rises above 40% for the wealthiest households. Second, realized capital gains are not representative of the broader stock of gains. Relative to realized gains, unrealized gains are more concentrated at the top of the wealth distribution and more heavily embedded in private businesses. For example, unlisted stock accounts for 91% of the observed stock of UCG in taxable asset categories, but only 56% of observed RCG in the same categories. Standard tax data on realized gains therefore make capital gains look more liquid, less private-business-intensive, and less top-heavy than the underlying stock of unrealized gains. Third, step-up exempts a large flow of gains embedded in highly appreciated inherited assets. In 2004, the last full year before step-up's removal was announced, inherited capital gains (exempted from tax under stepped-up basis) equaled 60% of the annual realized capital gains tax base. This large exemption arises because assets transferred to heirs contain far

[2] The United States briefly adopted carryover basis in 2010, but estates of Americans who died that year were ultimately allowed to choose between paying the estate tax and receiving stepped-up basis, or receiving carryover basis treatment. Gordon et al. (2016a,b) evaluate what may be learned from this brief, "voluntary" tax regime. The US also passed legislation that repealed step-up in favor of carryover basis in 1976, but implementation was delayed and the change was eventually repealed before it came into effect. See Gutman (2021) for a discussion of the administrative issues that motivated the repeal of the 1976 reform.

[3] In Section 3, we describe relevant institutional information, including various elements of taxation in Norway.

more embedded appreciation on average than assets their owners choose to sell: the ratio of capital gains to value is 75% for inherited taxable assets, compared to 6% for realized taxable assets. Fourth, the gains exempted by step-up accrue overwhelmingly to the top of the wealth distribution. The top 1% of households are the source of roughly two-thirds of inherited capital gains in taxable assets, and the top 0.01% alone is the source of roughly one-third.

In the second part of the paper, we examine how the removal of step-up in Norway affected realization behavior. Prior to 2006, Norway allowed accrued capital gains to disappear from the tax base when appreciated assets were gifted or bequeathed (a policy of "stepped-up basis" for both *inter-vivos* gifts and testamentary bequests). In 2006, Norway began "carrying over" the original owner's unrealized capital gains to recipients of stock (including private business interests and mutual fund shares) transferred as gifts or bequests. The reform did not change the capital gains tax rate faced by the original owner, but instead increased the future tax cost of leaving appreciated stock unrealized for intergenerational transfer. We present a conceptual framework showing that exposure to this cost increase is greater for older investors and for investors with a larger ratio of unrealized gains to wealth.

Our empirical strategies exploit these dimensions of differential exposure to the reform. We first estimate difference-in-difference regression models that compare the effect of the reform on the realizations of individuals aged 30–49 versus those of older individuals. The identifying assumption for this strategy is that, absent the reform, different age groups would experience the same percent change in average RCG over time. We find that investors in older age groups responded to the removal of step-up by increasing their taxable realizations relative to this counterfactual. Investors aged 50 and above increased yearly realized capital gains by $1,068 (36%) on average after the reform announcement, with response magnitudes that increase by age. As a second, alternative, empirical strategy, we compare realization responses by portfolio composition for investors of similar age and wealth. We estimate difference-in-difference regression models that compare the effect of the reform for investors with different levels of unrealized capital gains in stock prior to the reform. The identifying assumption is that, absent the reform, investors with different unrealized capital gains but the same age and wealth level would have capital gains realizations (of any asset type) that trend in parallel. Results from this design imply a similar increase in realized capital gains among older investors as in the first design. Increases in taxable realizations were largest among the investors with the most highly appreciated stock portfolios. The announcement also triggered a large anticipatory wave of *inter-vivos* transfers, but the simultaneous increase in taxable realizations shows that early transfers were an imperfect substitute for retaining appreciated assets.

Combining the two parts of the paper produces findings that cut in two directions. The removal of step-up has a large effect on the annual capital gains tax base, raising capital gains tax revenue by 17% according to our baseline estimates. However, the additional realizations only amount to 0.6% of the 2004 stock of unrealized capital gains per year, while new gains accrue at a much faster rate. Even after step-up is removed, the stock of unrealized capital gains continues to grow throughout our post-period. Removing step-up therefore captures revenue from gains that would otherwise have been inherited tax-free, but the stock of unrealized appreciation is so large that the reform does not by itself materially reduce it.

Step-up policy has been subject to debate in both the US and UK. Our results clarify that step-up benefits very wealthy households and that, relative to other policies that would benefit a similar group, step-up creates a negative fiscal externality. The reason is that step-up does not merely reduce taxes on inherited gains; it increases the return to deferral and thereby amplifies the lock-in distortion created by realization-based capital gains taxation. Using a "Marginal Value of Public Funds" framework, we estimate that step-up provides at most $0.80 in benefits for each $1 of foregone tax revenue. Step-up is therefore an unusually costly way to deliver tax benefits to capital gains owners: it transfers resources to a highly concentrated and wealthy group while increasing, rather than reducing, realization distortions.

Our paper contributes to two literatures. First, our data work complements a recent literature which seeks to incorporate yearly flows of accruing capital gains, asset returns or retained business income into a broader concept of personal income (Thoresen et al., 2012; Wolfson et al., 2016; Fairfield and Jorratt De Luis, 2016; Eika et al., 2020; Larrimore et al., 2021; Yagan, 2023; Alstadsæter et al., 2023; Aaberge et al., 2024; Fagereng et al., 2024).

Because prior papers do not link owners' original purchase prices to their contemporary holdings, they cannot measure the accumulated *stock* of unrealized gains or the gains embedded in intergenerational transfers. By reconstructing basis at the individual-asset level, we provide, to our knowledge, the first asset-level measures of unrealized and inherited capital gains in non-survey data.[4] Second, we add to a long literature on behavioral responses to capital gains taxation (e.g., Feldstein and Yitzhaki (1978)). Much of this literature estimates how realized capital gains respond to the capital gains tax rate, with recent estimates implying modest long-run realization elasticities.[5] We study realization responses to a different policy lever: changes to the tax treatment of inherited capital gains, holding the statutory rate fixed.[6] We find this margin to be quantitatively powerful: producing a comparable realization response through a rate cut would have (from back-of-the-envelope calculations using the elasticities of Agersnap and Zidar (2021) and Dowd and McClelland (2024)) required cutting Norway's capital gains tax rate by 5-12 percentage points. On the other hand, we contrast this "large" realizations response (the typical focus of the literature) with the small effect on the overall stock of unrealized capital gains.

The paper proceeds as follows: Section 1 describes our data, including how we create our novel measure of unrealized capital gains. Section 2 uses our novel data to document several new facts about unrealized and inherited capital gains in Norway under the stepped-up basis regime. Section 3 provides background on Norway's policy change from stepped-up basis to a carryover basis regime. Section 4 provides a conceptual framework for understanding how the tax treatment of inherited capital gains (e.g., stepped-up basis) affects investors' realization decisions. Section 5 examines how the removal of step-up affected taxable realizations and intergenerational transfers in Norway. Section 6 discusses policy implications and the

[4] The US Survey of Consumer Finance (SCF) measures unrealized capital gains — see, e.g., Looney and Moore (2016). Unfortunately, the SCF, by design, excludes the very wealthiest US households, preventing a complete characterization of the distribution of unrealized capital gains. The SCF's cross-sectional structure also makes it difficult to use to analyze behavioral changes over time. At the aggregate level in Norway, Andresen (2022) estimates the sum of all unrealized capital gains in unlisted stock. We compare our estimates to Andresen (2022) in Appendix D.6.

[5] Agersnap and Zidar (2021) estimate elasticities implying revenue-maximizing rates of 38-47 percent, and Lavecchia and Tazhitdinova (2024) find small permanent realization responses to the cancellation of a lifetime exemption in Canada.

[6] Although we provide the first quasi-experimental evidence examining a permanent change to the tax treatment of inherited capital gains, we are certainly not the first to theorize that step-up likely depresses capital gains realizations (Holt and Shelton, 1962; Auerbach, 1989; Kiefer, 1990).

generalizability of our results.

1 Data and Measurement

Our paper uses data and policy variation from Norway, primarily because Norway is the only country that has enacted a permanent change to its stepped-up basis policy in recent decades.[7] Norway is also a uniquely appropriate setting for this project because the Norwegian Tax Authority collects unusually rich data on taxable income, wealth, and asset transactions.

The core data contribution of this paper is an asset-level reconstruction of the capital-gains universe. Throughout the paper, we refer to the stock of unrealized capital gains as UCG, and the flows of realized and inherited capital gains as RCG and ICG, respectively. Norwegian tax returns directly measure RCG, but neither tax returns nor standard wealth records measure the stock of UCG or the ICG embedded in assets transferred to heirs. We construct these measures by linking transaction records, end-of-year holdings, purchase prices, current asset values, and transfer types at the individual-asset-year level. Appendix Table A1 summarizes the data sources, coverage periods, units of observation, and the role each source plays in our measurement exercise.

As background, Norway is a small, open economy with a well-educated population, a sizable middle class, and a significant public sector. The Norwegian population is 5.4 million people and consistently ranks among the wealthiest global populations.[8] Levels of wealth inequality are relatively high, and top wealth is concentrated in real estate and private business interests (see Appendix A). Norwegian administrative data is considered very high quality and much of the information (nearly all data on income and financial wealth) is third-party reported from employers, banks, and financial intermediaries. Throughout this paper, we convert all currency amounts to 2018 USD using yearly average consumer price indices from Statistics Norway and the NOK-USD exchange rate on December 31, 2018.

[7] See Appendix Section F for background on stepped-up basis policy from a global perspective.

[8] In 2023, GDP per capita in Norway was \$87,961, versus \$81,695 in the United States (World Bank, 2023).

1.1 Administrative Data Sources

The raw data underlying our measurement exercise combine two broad types of Norwegian administrative records: annual income and wealth tax records, and asset-level transaction records.

First, as in many countries, Norwegian taxpayers report taxable RCG as a component of income on their yearly tax returns. We use this aggregated RCG variable from 1998 to 2018 as our primary outcome variable in our regressions in Section 5. For a subset of years, we can decompose aggregate taxable RCG into taxable RCG from real estate, from unlisted stock and from other asset types (primarily listed stock and mutual funds).[9] The annual tax records also include detailed information about each Norwegian household's net wealth. The Norwegian wealth data are perceived to be high quality, and we are confident that they capture market values well for most components of financial wealth, such as bank deposits, liabilities, and listed securities. For unlisted companies we replace the tax-assessed values with our estimates of market value described in Section 1.2.2. We also replace tax-assessed real estate values with property-level values computed by Eika et al. (2020).[10]

Second, since the Norwegian Tax Authority does not collect data on UCG or ICG, we use asset-level transaction data to build these measures ourselves for the asset categories of Norwegian stock (including private businesses) and real estate. We observe individual transactions, including sales, inheritance transfers and other transaction types, for listed and unlisted stocks[11] over the period 2003 to 2018. These data contain detailed information on the type of transfer, the identity of the buyer or seller, and the transaction price. The data contain information on the last transaction prior to January 1, 2003, and all subsequent transactions from 2003 to 2018. We have similar real estate transaction data covering 1993 to 2018 (including the last transaction prior to 1993) from the Norwegian Land Register.

[9] Specifically, the tax returns allow us to decompose aggregate RCG into real estate and non-real estate RCG for all years except 2005 (due to a data corruption). Once the stock transactions data starts in 2003, we can further decompose non-real estate RCG into RCG from and not from particular kinds of stock.

[10] Although the Norwegian tax authority assesses the value of all domestic real property each year, the quality of the assessment methodology prior to a substantial revision in 2010 is thought to be poor, and the 2010 reform did not improve the quality of the assessments for non-residential property. The Eika et al. (2020) values we use for real estate valuation are built from the same real estate transaction data we use to track ownership by interpolating and extrapolating from sale values with house price indices in years when the specific property does not sell.

[11] Our category of unlisted stock includes all private business interests other than sole proprietorships.

The pre-period transaction information allows us to initialize holdings and basis for assets already held when the panel begins. For example, if an individual bought shares in 1985 and continued to hold them in 2003, we observe both the quantity held and the original purchase cost.

1.2 Constructing Asset-Level UCG and ICG

At the individual-asset level, capital gains are measured as the current value of the asset minus the owner's purchase cost, which is called the owner's "basis" in the asset. That is, Capital Gains = Current Value - Basis. For RCG, the current value in this expression is observed directly as the transaction price when the asset is sold or otherwise realized. The measurement challenge in this paper is that an analogous current value is not directly observed when an asset remains unsold at a particular point in time, or when it is transferred to heirs without an arm's-length sale price. We therefore need to reconstruct these values for our UCG and ICG estimates.

1.2.1 Tracking Basis and Current Holdings

To construct our capital gains measures, we first convert the transactions data into a ledger at the individual-asset level. When an individual buys a new asset, we open a ledger for that individual-asset pair. We record the number of shares initially owned in the asset (or fractional ownership share for real estate) and then update the number of shares held (or fractional ownership held) as the individual makes subsequent transactions in the asset. We account for stock splits, consolidations and mergers when applicable.

We measure an individual's initial basis in an asset as the asset's original purchase cost, or carryover basis if applicable. To properly account for carryover basis, it is necessary to trace assets moving across ledgers in non-taxable transactions. Basis may be carried over between ledgers due to mergers, changes to share classes, or intergenerational transfers after the removal of step-up (The Norwegian Tax Administration, n.d.). As the individual buys and sells, we increase and decrease his basis under a first-in-first-out assumption, in keeping with the accounting methods required by Norwegian law.[12]

[12] Appendix Section D.3 shows that our measure of UCG is qualitatively unaffected if we instead assume

1.2.2 Measuring Current Value for Unrealized Capital Gains

To construct UCG, we require a yearly measure of each asset's market value, even when that asset is not sold. For listed stock, we obtain the stock's end-of-year closing price from Thomson-Reuters Datastream. For unlisted stock, we begin with the book value measure used by the Norwegian government to value each unlisted company for wealth and inheritance tax purposes.[13] Valuing private businesses is inherently difficult, and tax-assessed book values are likely to understate market values for many unlisted firms. We therefore create our own measure of unlisted company value by scaling up the company's sales, income, and asset book values with multipliers computed for a sample of listed firms in the same industry, and then applying a 10% liquidity discount to those values. This procedure follows Smith et al. (2023) and is described in greater detail in Appendix D.2. When a firm's estimated value is negative, we replace that value with zero, in keeping with the limited liability of the firms considered. When unlisted firm value is directly attributable to firm ownership in another firm or listed equity, we value that portion of the firm as the pro-rata share of the child firm's value or using the value of the listed equity. As our yearly measure of real estate value, we again use the property-level values computed by Eika et al. (2020).

1.2.3 Measuring Current Value for Inherited Capital Gains

We observe gift/inheritance transfers of stock and real estate. To construct our measure of ICG, we assign the relevant portion of the predecessor's basis to the transferred units and calculate ICG as the value of the transferred asset minus the basis associated with the asset. For listed stock, we value transferred shares using observed market prices at the time of transfer. For unlisted stock and real estate, inheritance transactions do not usually include a recorded market value, so we use the same beginning-of-year valuation convention as in our UCG construction. Thus, for these assets, ICG equals the estimated beginning-of-year value of the transferred asset minus the basis associated with the transferred units.

capital gains are realized according to an average basis assumption.

[13] When stepped-up basis was in place, this valuation also served as the heir's new basis for unlisted stock received in inheritance.

1.2.4 Validation

Our capital gains measures rely on reconstructing holdings and basis from transactions. Appendix Section D validates our ledger-implied holdings against end-of-year shareholder reports in the Norwegian Shareholder Registry, available for limited liability companies since 2004, and finds that our ledger agrees very closely with the Registry: an exact match for 97% of current shareholders. We also validate our measure of basis using realized transactions. For capital gains that are eventually realized, we can calculate realized capital gains as the recorded value of the sale minus basis and compare that calculation to the realized capital gains reported in each individual's tax return. This match is also good — 97% of individual yearly tax returns agree, within rounding error, with our calculation for the amount of realized capital gains in stock (see Appendix D).

1.3 Coverage and Limitations

We focus on capital gains in what we call *taxable asset categories* — that is, asset categories that are usually subject to the capital gains tax.[14] Taxable asset categories in our data include listed and unlisted stock, mutual funds, and taxable real estate. Primary residences and certain other forms of real estate, although present in our data, are exempt from capital gains tax if the owner meets certain holding period requirements.[15] These categories of *non-taxable real estate* do not typically benefit from stepped-up basis, and consider their capital gains separately in the analysis that follows.

We directly reconstruct UCG and ICG for Norwegian stock (listed and unlisted) and real estate. A limitation of the transaction data is that we do not have access to transactions in mutual funds, which we know from aggregate statistics to be responsible for nearly all RCG that are not from directly owned stock or real estate (Statistics Norway, 2007). Since we observe wealth in mutual funds but cannot calculate their UCG, we bound our estimates

[14]The use of the term "taxable asset categories" is not meant to imply that the capital gains within these assets will necessarily be subject to the capital gains tax, only that they would be so subject if realized. Capital gains from retained earnings might instead be paid out as dividends, and thus subject to a dividend tax (if any). And, in accordance with the focus of this paper, capital gains passed in intergenerational transfers under step-up are exempted from the capital gains tax base.

[15]Appendix Section F details the holding period requirements for the real estate exemptions from capital gains taxes.

of total UCG by making the alternative assumptions that (1) Norwegian households do not have any UCG in mutual funds and (2) Norwegian households' mutual fund wealth is 100% UCG, i.e., all owners of mutual funds have zero basis in their mutual fund holding. In practice, the treatment of mutual funds does not qualitatively affect any of our results. While mutual funds are an important component of many Norwegians' portfolios, they are much less important for the richest Norwegians (see Appendix Figures A6 and A7). We do not directly measure UCG in other asset classes, such as directly held international stock, directly held business assets, and artwork. However, the aggregate amount of wealth in these categories, and thus the upper bound for the possible amount of UCG, is small.[16] See Appendix Figures A6 and A7.

A few categories of wealth are missing. We do not observe wealth in defined contribution pension plans. Defined contribution plans were not permissible in Norway until 2001 and only begin to have market share during our period of study.[17] Although Norwegians are legally required to report wealth both inside and outside Norway, we do not observe wealth that Norwegians fail to report to the tax authority. Alstadsæter et al. (2019) establish that offshoring forms of tax evasion are substantial among the wealthy citizens of Norway. Our inability to observe this offshore wealth may mean our estimates understate the amount of UCG in portfolios at the top of the wealth distribution or incorrectly rank evading households in the wealth distribution. On the other hand, since these assets are, by definition, hidden from the Norwegian tax system, we do not expect realizations or inheritance of these assets to affect our estimates of the step-up reform.

2 Step-Up in Context: The Magnitude and Distribution of Capital Gains

In this section, we use our novel data to document the magnitude, composition and distribution of unrealized and inherited capital gains under step-up (i.e., prior to the Norwegian policy reforms). This section focuses on capital gains in Norway as of end-of-year 2004,

[16]Most Norwegians participate in international equity markets via mutual funds, and this wealth would therefore be included in our mutual fund numbers.

[17]Even in 2017, the average wealth in individual and private contribution plans was only around $24,000 (Smogeli and Halvorsen, 2019).

immediately before the removal of step-up was announced.

2.1 Fact 1. Unrealized Capital Gains Are a Large Fraction of Overall Household Wealth

UCG in taxable asset categories are large: twelve to fourteen percent of total household wealth. See Table 1 Panel (a). At the end of 2004, \$64.85–\$75.05 billion of UCG were outstanding in asset categories subject to the capital gains tax: listed and unlisted stock, mutual funds, and taxable real estate.[18] This amount equals 12–14% of aggregate net wealth held by Norwegian households in 2004 (\$531 billion). Primary residences and certain other forms of real estate are typically exempt from the capital gains tax and are excluded from these totals.[19]

If we look only at the \$118.95 billion of wealth in taxable asset categories, 55–63% was UCG. Unlisted stock was particularly appreciated: 66% of wealth in unlisted stock was UCG, compared to 21% in listed stock and 41% in taxable real estate (Table 1 Panel (a)). These magnitudes establish the basic scale of the latent capital gains tax base: before considering which gains are eventually sold or transferred, accumulated untaxed appreciation is a quantitatively important component of household wealth.

2.2 Fact 2. Realized Gains Are Unrepresentative of Unrealized Gains

Our data allow us to compare RCG to the broader stock of UCG, and show that RCG are not representative of the distribution of unrealized gains. Relative to the RCG observed in tax data, the UCG stock differs along two key dimensions: its asset composition is more heavily

[18] We present our measures of total unrealized capital gains in taxable assets as a range to acknowledge our uncertainty regarding the amount of unrealized capital gains in mutual funds. We do not observe mutual funds in our transaction data and are therefore unable to measure the amount of UCG in mutual funds. However, we know that an individual's unrealized capital gains in mutual funds cannot exceed his total wealth in mutual funds, which we observe in our wealth data. We can therefore present a range for the value of unrealized gains including mutual funds: the upper bound of this range assumes that all mutual fund holdings are 100% appreciated. The lower bound of this range assumes zero unrealized capital gains in all mutual fund holdings.

[19] Appendix Section F details the holding period requirements for the real estate exemptions from capital gains taxes.

tilted toward private businesses, and its distribution across households is substantially more concentrated at the top of the wealth distribution.

Most realizations (87%) are from listed shares and RCG is attributable to a mix of asset classes.[20] On the other hand, private (unlisted) business assets are responsible for the great majority of wealth in capital gains-producing assets, UCG, inherited assets and ICG. Table 1 shows that unlisted stock accounts for 82% of value in our taxable asset categories ($89.80 billion of the $108.75 billion), 91% of the observed UCG stock, 83% of inherited assets and 87% of inherited capital gains. By comparison, unlisted stock accounts for 56% of observed RCG and 11% of realized asset value.

Figure 1 Panel (b) compares the concentration of RCG and UCG across households ranked by total net worth. The UCG curve lies below the RCG curve through the upper part of the distribution, indicating that UCG are more concentrated at the top than RCG. Put differently, tax data on RCG make capital gains look less top-heavy than when unrealized capital gains are included. In 2004, the bottom 90% of households owned less than 25% of the stock of taxable UCG in Norway. The next 9% of households owned 26%, and the top 1% of households owned 49–54% of the stock of UCG.[21]

The importance of UCG within household portfolios also varies across the wealth distribution. Figure 1 Panel (a) shows that taxable UCG are less than 10% of wealth for the bottom 80% of households in 2004. The ratio of taxable UCG to total household wealth increases sharply at the top of the wealth distribution, reaching a value of 47% for the wealthiest households.

The skewed distribution of taxable UCG is mostly driven by differences in household portfolio composition across the wealth distribution. Wealthy households hold large fractions of their wealth in assets subject to the capital gains tax — holdings of unlisted stock are

[20] Although listed shares make up 87% of realized asset value, realizations of listed stock frequently contain losses or very little capital gain. Although unlisted stock are realized less frequently, they tend to be much more appreciated (a 31% RCG/Amount ratio, versus 3% for listed stock) and therefore make up a larger share of RCG than of realized asset value. Note that our numbers in Panels (b) and (c) of Table 1 do not contain mutual funds, which we know from external data to be another important source of realized capital gains, but whose contribution to the stock of unrealized capital gains is bounded above at 14% (Table 1 Panel (a)).

[21] As before, we present ranges to incorporate alternative assumptions regarding the amount of unrealized capital gains within mutual funds.

particularly important for the wealthiest households. For example, households in the top 0.01% of net wealth in 2004 own 22% of the total wealth in stock and taxable real estate (see Appendix Figure A7). These top households own a similar share of unrealized capital gains in these asset categories. In contrast, less wealthy households own more of their wealth in cash and primary residences, which are not subject to the capital gains tax.

2.3 Fact 3. Step-Up Exemptions Are Large Relative to the Tax Base but Small Relative to the UCG Stock

Under step-up, ICG constitute a small but highly appreciated outflow from the stock of UCG and a large exemption relative to the annual RCG tax base. In 2004, \$550 million of capital gains in stock and taxable real estate were passed through inheritance and thus exempted from tax under the stepped-up basis policy. In the same year, \$910 million of capital gains were realized and taxed, from a total stock of approximately \$65 billion of UCG in stock and taxable real estate (Table 1 column 2). Therefore, Norway's stepped-up basis policy exempted an amount equal to 60% of the annual capital gains tax base.

The fact that only 1.65 times as many capital gains were realized as inherited in 2004 is striking because nineteen times more value was realized than inherited (Table 1, Panels (b) and (c), column 1). Inherited assets are much more highly appreciated than those that are realized. The ratio of taxable capital gains to value in inherited assets subject to the capital gains tax is 75%. Realized assets are much less appreciated: the ratio of capital gains to value in realized assets subject to the capital gains tax is 6%. Listed and unlisted stock show a particularly large difference in appreciation between inherited and realized assets: 69% in inherited versus 3% in realized listed stock, and 79% in inherited versus 32% in realized unlisted. Appreciation differences between realized and inherited real estate, whether within the taxable or non-taxable category, is less notable (Table 1, Panels (b) and (c), column 3). The appreciation gap explains why step-up exempts such a large fraction of the capital gains tax base despite applying to a small volume of transfers: inherited assets are systematically the most appreciated assets in the economy.

However, these benchmarks reveal a nuanced picture of step-up's significance. Relative to the capital gains tax base, step-up is large, so removing step-up could be expected to have a substantial mechanical effect on capital gains tax collections. But relative to the stock of UCG, both RCG and ICG are small yearly outflows from accumulated appreciation: in 2004, ICG in stock and taxable real estate equaled 0.8% of the UCG stock in those categories, while RCG equaled 1.4% of the same stock.

2.4 Fact 4. The Inherited Gains Exempted by Step-Up Come Overwhelmingly from the Top

The benefits of stepped-up basis depend on whose capital gains are passed through inheritance. ICG are the gains directly exempted by step-up, and these gains are highly concentrated at the top of the wealth distribution. Figure 1 Panel (b) shows that the bottom 90% of households are the source of only 14% of ICG in taxable assets in 2004. The next 9% of households are responsible for 20% of ICG, while the top 1% is the source of the remaining 66%. The top 0.01% alone is responsible for 34% of ICG in taxable assets.

Figure 1 Panel (c) translates these distributional facts into their implications for personal tax burdens. We calculate the mechanical change in average personal taxes across the wealth distribution under two counterfactual scenarios: one in which RCG are exempted from tax, and one in which ICG are also taxed at the 28% capital gains tax rate. Both capital gains categories have a near-zero effect on personal taxes for most of the wealth distribution. But in the upper percentiles, removing step-up would mechanically increase personal taxes by approximately 10% for the very wealthiest households. This is meaningful—but bounded, and similar in size to the approximately 12% decrease in personal taxes at the top that would result from exempting all RCG. The comparison also illustrates the relative concentration of ICG versus RCG: the taxation of RCG affects a broader swath of the upper distribution, while ICG matter only at the very top.

Having established that step-up is an important and regressive feature of the capital gains tax system—but one whose effects on the distribution of total taxes are limited—we now turn to examining how investor behavior changed when step-up was removed. In the sections that follow, we describe the Norwegian policy change (Section 3), develop a conceptual framework for understanding the behavioral response (Section 4), and estimate the magnitude of the realization response (Section 5).

3 Norway's Removal of Step-Up for Stock (2005-2006)

The remainder of this paper studies the removal of stepped-up basis for stock in Norway (including listed and unlisted stock, and equity-based mutual fund shares). This change was announced in 2005 and implemented in 2006. Capital gains and inheritance tax rates did not change during the same period. Further detail on later policy changes, including the removal of step-up for non-stock asset classes (concurrent with the removal of the inheritance tax in 2014), can be found in Appendix F.

During most of our period of study, capital gains realized by their original owner were taxed at a 28% rate.[22] Heirs were subject to an inheritance tax on the total value of received gifts and bequests (of any type), but, until 2006, were not subject to capital gains tax on capital gains accrued during their predecessor's holding period. The top inheritance tax rate for transfers between close relatives was 20% from 1999 to 2008.[23] We end our short-term difference-in-difference results in 2008 to isolate the removal of step-up for stock from subsequent changes to inheritance tax rates (beginning in 2009) and capital gains tax rates (beginning in 2014)..

The repeal of step-up was enacted as a part of the broader 2006 shareholder-tax reform. (see Appendix Section F.1.4 for more detail about the reform, and Section 5.3.2 for a discussion of the implications for our identification strategy). Although the main components of the 2006 tax reform were formally announced in March 2004 and approved in June of that year, the removal of step-up was not part of the June legislation and was not definitively decided until December 2004 (Norwegian Ministry of Finance, 2004). Since we do not find any press coverage regarding the impending removal of step-up until January 2005 (see Ap-

[22] The capital gains tax rate was 28% from 1992 to 2013 and has never depended on individual income (i.e., there is no progressive rate structure). The same rate was also applied to other forms of capital income, such as interest income. Net capital losses are fully deductible against ordinary income at the capital gains tax rate (so a $1 net loss implies $0.28 in tax relief).

[23] Norway's inheritance tax had a rate structure that depended on the cumulative amount of the inheritance received, and the relationship between the giver and the recipient. Other than a small yearly gift exemption, no distinction was made between *inter vivos* gifts and bequests at the time of death. Appendix Figure A1 and Table A2 provides details on the rates and brackets over time.

pendix F), we take 2005 to be the relevant announcement year when considering anticipatory responses in our empirical analysis.

The removal of step-up for stock meant that, from January 1, 2006, intergenerational transfers of appreciated stock (including unlisted stock and equity-based mutual funds) received carryover basis treatment.[24] That is, heirs who inherited appreciated stock continued to be subject to inheritance tax on the value of the stock they received, but now were also subject, if and when they sold the stock, to capital gains tax on the amount of appreciation during the original owner's holding period.[25] Losses, however, were not permitted to be carried over in the 2006 reform, so stocks with losses continued to be subject to a "step-down" in basis at the time of inheritance. Step-up also continued for non-stock assets (such as real estate) until 2014.[26]

Our empirical analysis focuses on how stock owners reacted to the 2005 announcement of step-up's impending removal, rather than the later implementation of that removal. The announcement date is important because the capital gains tax treatment of the original owner is not changing. The original owner faces the same cost of realizing his capital gains before or after step-up is removed. But the announcement of step-up's removal informs the original owner that the inheritance he may leave in the future will be more expensive, which will cause him to reassess his trade-off between present realizations and future intergenerational transfers. We formalize this understanding of step-up's effects on owners of unrealized gains in the next section of the paper.

Although repealing stepped-up basis and inheritance taxes are frequently considered in tandem, our empirical setting allows us to isolate the effects of removing step-up from any changes to the inheritance tax regime. Conceptually, an inheritance tax is not a perfect substitute for exempting inherited capital gains from the capital gains tax base. Inheri-

[24]In Norway, mutual fund investors are not taxed when fund managers buy and sell stocks or reinvest dividends. Tax is only triggered when the investor himself sells his mutual fund units or when the investor receives taxable distributions.

[25]From 2006 until the removal of the inheritance tax in 2014, recipients of gifted or inherited stock were given a deduction on their inheritance tax bill equal to 20% of the carried-over capital gains, to lessen the "double taxation" from being subject to both an inheritance tax and carried over capital gains. We abstract from this deduction in our explanation of the policy changes, but fully account for it in all our figures and empirical analyses.

[26]See Appendix F for detail on the removal of step-up for non-stock assets, which is not the focus of our empirical analysis.

tance transfer taxes apply to the current value of the bequest, regardless of any unrealized capital gains. Under stepped-up basis, an investor who sells a highly appreciated asset the day before death and leaves the resulting cash as inheritance is subject to both the capital gains tax and the inheritance tax, whereas a dying investor who bequests the unrealized asset itself pays only the inheritance tax (if any). As our conceptual framework in Section 4 demonstrates, three distinct policy levers affect the trade-off between capital gains realizations and intergenerational transfers: the capital gains tax rate, the inheritance tax rate, and step-up.[27]

4 Conceptual Framework

This section presents a lifecycle model with taxes on capital gains realizations and bequests. Our model follows a broad literature on lifecycle portfolio models (surveyed in Gomes (2020)) and is more specifically an extension of the model in Dammon et al. (2001). We use this model as a framework to understand the effects of removing step-up. The removal of step-up reduces the after-tax value of an inheritance in the hands of the heir. Under the assumption that givers of intergenerational transfers have altruistic bequest motives (i.e., that they care about the taxes their heirs pay on the inheritance), we show that substitution effects from this price change will encourage increased taxable realizations during the original owners' lifetimes. We also simulate the model (it is not solvable analytically) in order to discuss dynamics and heterogeneity in the expected response.

4.1 Lifecycle Problem

Consider an individual who may live from time $t = 0, \dots T$, where $\mathcal{T} \in \{1, ..., T\}$ denotes the stochastic period of death. Death occurs by period T with certainty and mortality risk is exogenous and independent of asset price shocks. The individual's wealth at the beginning of period t is W_t. This wealth is partially held in the form of a risky asset: the individual

[27] There is a small existing literature on the interactions between inheritance and capital gains taxation. Prior work illustrates two of the three "policy levers" mentioned above: Auten and Joulfaian (2001) show that higher estate tax rates encourage more capital gains realizations by the original owner. Poterba (2001) provides evidence that larger unrealized capital gains push owners towards the utilization of stepped-up basis. We are the first paper to show empirically that changing stepped-up basis affects taxable realizations.

owns n_{t-1} shares priced at P_t at the beginning of time t. These shares contain an amount of unrealized capital gains U_t at the beginning of period t. The individual holds the rest of his wealth in a risk-free asset earning an (after-tax) nominal return of r. In each period $t < \mathcal{T}$, the individual chooses after-tax consumption (c_t) and shares in the risky asset to hold into next period (n_t). The individual faces capital gains tax (τ_c) on realized gains when he sells risky assets. The price of the risky asset changes between periods and wealth and unrealized capital gains evolve accordingly. In period $\mathcal{T}$, the individual dies and remaining wealth passes to his heir as a bequest (with post-tax value $B_{\mathcal{T}}$). Heirs face inheritance tax (τ_I) on the value of any bequest. When there is a carryover basis regime (denoted with the policy indicator $\theta = 1$), the heir will also have to pay capital gains tax (τ_c) to realize their predecessor's terminal unrealized capital gains.

The individual maximizes utility subject to per-period budget constraints and a function that transforms wealth at death into a post-tax bequest amount:[28]

$$\begin{aligned}
&\max_{\{c_t, n_t \geq 0\}_{t=0}^{\mathcal{T}-1}} \quad \mathbb{E}_0 \left[\sum_{t=0}^{\mathcal{T}-1} \beta^t u(c_t) + \beta^{\mathcal{T}} \nu(B_{\mathcal{T}}) \right] && s.t. \\
W_{t+1} &= \big(1+r\big)\big(W_t - n_t P_t - c_t - \tau_c R_t\big) + n_t P_{t+1} && \forall t = 0, ...T-1; \\
B_t &= W_t\big(1-\tau_I\big) - \theta \tau_c U_t^+ && t = \mathcal{T};
\end{aligned} \tag{1}$$

where r is the nominal return on the risk-free asset, $U_t^+ \equiv \max\{U_t, 0\}$ (in keeping with the fact that the Norwegian policy reform only carried over gains) and R_t is capital gains realized at time t. The fraction $g_t \equiv R_t/U_t$ of beginning-of-period unrealized capital gains that are realized at time t depends on whether there is an embedded gain or loss in the beginning-of-period shares and whether the investor chooses to buy or sell shares this period:

$$g_t \equiv \begin{cases} \max(1 - \frac{n_t}{n_{t-1}}, 0) & \text{if } U_t > 0 \quad \textit{(embedded gain)} \\ 1 & \text{if } U_t \leq 0 \quad \textit{(wash sale on embedded loss)} \end{cases}$$

We assume that all shares will be sold and immediately repurchased whenever there is an

[28]Note that our setup assumes the heirs realize their inheritance immediately and that the giver cares about the post-tax bequest amount.

embedded loss ($U_t < 0$).[29] Under an average basis rule, unrealized capital gains evolve according to:[30]

$$U_{t+1} = (P_{t+1} - P_t)n_t + \max(U_t, 0)\frac{n_t}{\max(n_t, n_{t-1})} \tag{2}$$

where the first term represents new appreciation from price changes in period $t+1$ and the second term represents accumulated appreciation from prior periods (if any).

4.2 Realizations under Step-Up versus Carryover

We now use the model to characterize how the tax treatment of inherited capital gains affects realization choices during the original owner's lifetime. Under step-up ($\theta = 0$), the heir's basis is reset and the predecessor's terminal embedded gains $U_{\mathcal{T}}$ do not generate a tax liability for the heirs. Under carryover ($\theta = 1$), the heir inherits the predecessor's basis and therefore inherits latent capital gains tax liability on positive terminal gains $U^+_{\mathcal{T}}$.[31]

4.2.1 Baseline Prediction: Increased Realizations under Carryover

We can rewrite the giver's problem for state variables $X_t = [P_t, U_t, n_{t-1}, W_t]'$ as

$$V_t(X_t;\theta) = \mathbb{P}(\mathcal{T} > t|\mathcal{T} \geq t) \max_{c_t, n_t \geq 0} \big\{u\big(c_t\big) + \beta E_t[V_{t+1}(X_{t+1};\theta)]\big\} + \mathbb{P}(\mathcal{T} = t|\mathcal{T} \geq t)\nu\big(B_t\big) \quad s.t.$$

$$W_{t+1} = \big(1+r\big)\big(W_t - n_t P_t - c_t - \tau_c R_t\big) + n_t P_{t+1}$$

$$B_t = W_t(1-\tau_I) - \theta\tau_c U^+_t$$

It is clear that the removal of step-up ($\Delta\theta$) affects the giver only through the payoff in the event of death, $\nu(B_t)$. The change in this death payoff from a marginal policy move from

[29] Without transaction costs or wash sale restrictions, investors with losses will optimally liquidate and then immediately repurchase all shares. Since realizations in the model arise only from net reductions in shareholdings, sale-and-repurchase of gain positions is not in the choice set.

[30] Under this formula, the tax basis for currently held shares is the weighted average purchase price of those shares. This is an average basis assumption and is necessary for tractability in models with more than a few periods since a (more realistic) first-in-first-out basis assumption requires separate state variables for each asset purchase (Dybvig and Koo, 1996). DeMiguel and Uppal (2005) show, in a model broadly similar to our own, that the certainty equivalent loss from using the average tax basis instead of the exact basis is small. Our empirical results are also robust to calculating unrealized capital gains under either basis assumption (Appendix Figure A8).

[31] The use of $U^+_{\mathcal{T}}$ reflects Norway's 2006 asymmetric carryover reform: terminal gains but not terminal losses were carried over to heirs.

step-up towards carryover is:

$$\frac{\partial \nu(B_t)}{\partial \theta} = -\tau_c U_t^+ \cdot \nu'\big(W_t(1-\tau_I) - \theta\tau_c U_t^+\big) \leq 0$$

This is the utility cost of the policy change to the giver if he dies in period t. Since ν is increasing (givers value bequests), removing step-up hurts givers with $U_t > 0$ and has no effect at death when $U_t \leq 0$. Effectively, the switch to carryover increases the price (i.e., reduces the post-tax value) of bequests relative to lifetime realizations for the giver. If substitution effects dominate income effects along this margin, we expect the giver to choose a weakly higher level of taxable realizations under carryover than under step-up.[32]

4.2.2 Heterogeneity in Exposure (Comparative Statics)

The following discussion characterizes how exposure to (i.e., current period utility costs for) the removal of step-up differ across investors. If substitution effects dominate income effects, these exposure gradients imply heterogeneous treatment effects of the reform on lifetime realizations, which we exploit in our empirical strategies in Section 5.

Remark 1: Exposure by Age. Conditional on his subjective distributions over (W_s, U_s) at possible death dates s, the closer an investor is to his time of death the more the removal of step-up matters for his current choices. Formally, for an investor alive and making choices at time $t < \mathcal{T}$,

$$\frac{\partial V_t}{\partial \theta} = \sum_{s=t+1}^{T} \beta^{s-t} \cdot \mathbb{P}(\mathcal{T} = s | \mathcal{T} > t) \cdot \mathbb{E}_t\left[\frac{\partial \nu(B_s)}{\partial \theta}\right]$$

which follows from the envelope theorem applied to the Bellman equation, since θ only enters the death payoffs of the Bellman equation. The weights $\beta^{s-t}\mathbb{P}(\mathcal{T} = s|\mathcal{T} > t)$ show the age channel: holding the expected death-state derivative fixed across remaining horizons, a shift in mortality toward earlier dates (as happens when age rises) makes the policy's effect less heavily discounted.[33] All else equal, older investors are therefore more exposed to the

[32] In order to translate these price changes into behavioral predictions, we need to assume that substitution effects dominate income effects along the realization-bequest margin and that the magnitude of the substitution effect is monotone in the size of the price change. These assumptions will be implicitly maintained by the parameterization of the model in our numerical simulations.

[33] Age also shifts the death-payoff terms $\mathbb{E}_t[\partial \nu(B_s)/\partial \theta]$ themselves, since investors who die later have had more time to accumulate wealth and embedded gains; Remark 1 isolates the mortality-timing channel by

removal of step-up. In our analysis of the removal of step-up for stock in Norway in Section 5, this observation guides our primary age-based empirical strategy.

Remark 2: Exposure by Unrealized Gains at Death. Holding age and beliefs about W_s fixed at each possible death date s, investors who expect to die with more taxable unrealized capital gains are more exposed to the removal of step-up. Differentiating the death payoff at $\theta = 0$,

$$\frac{\partial \nu(B_s)}{\partial \theta}\Big|_{\theta=0} = -\tau_c U_s^+ \cdot \nu'(W_s(1-\tau_I))$$

shows that the welfare cost is weakly increasing in U_s at each W_s (strictly so on gains). Formally, a first-order stochastic increase in $U_s \mid W_s$ at each possible death date therefore raises the magnitude of $\partial V_t/\partial\theta$.[34] Investors who expect to die with more embedded gains at a given terminal wealth level lose more from the removal of step-up and therefore (conditional on age and wealth) have a stronger substitution effect towards lifetime realizations. In our analysis of the removal of step-up for stock in Norway in Section 5, this observation guides our second empirical strategy exploiting heterogeneity across portfolio positions.

These comparative statics characterize the impact effect of the policy change holding the investor's portfolio composition fixed. In the medium and long run (discussed further below), the distributions of U_s, n_s, and W_s at possible death dates are endogenous to the policy regime, as investors optimally adjust their portfolios under carryover.

4.3 Lifecycle Realizations under Step-Up and Carryover

Thus far, we have focused on impact exposure to the removal of step-up: we have examined comparative statics holding investor portfolios fixed. Our empirical analysis, however, will exploit a change in policy which will induce both short-term (transitional) and long-term (equilibrium) responses to the removal of step-up. This subsection compares realizations when one or another of the two policy regimes is always in place, deferring further discussion of the transition dynamics between them to the empirical section. For these simulations, we parameterize the model with $T = 80$, mortality following Norwegian life tables, isoelastic

holding these fixed across horizons. The simulations solve the full model and capture both channels.

[34] Since $|\partial\nu(B_s)/\partial\theta|$ is non-decreasing in U_s at each W_s, a conditional FOSD shift in $U_s \mid W_s$ raises $\mathbb{E}_t[|\partial\nu(B_s)/\partial\theta|]$ by iterated expectations on the fixed marginal of W_s.

utility (implying choices scale with wealth) and an exogenous binomial process for the price evolution of the risky asset. Further details and parameters for the model simulation are described in detail in Appendix B.

Both the level and shape of capital gains realizations over the lifecycle differ between the step-up and carryover policy regimes. In the step-up regime, individuals hold a larger share of their portfolio in equity throughout the lifecycle and avoid capital gains realizations late in life. In the carryover regime, equity is drawn down as individuals age, implying more taxable realizations late in life (Figure 2 Panel (a)). Therefore, moving from a step-up regime to a carryover regime entails two types of realization responses: (i) in the short term, people who built up equity shares that are higher than optimal for the carryover regime need to reduce their equity holdings, which requires taxable sales and (ii) the carryover regime has persistently higher levels of realizations. Both of these effects are stronger for older individuals. The short-term effects are increasing in age because older people have a larger gap between optimal equity holdings under the two policy regimes (and thus a larger need to sell). The long-term effects are increasing in age because older people in the carryover regime realize more.

In the next section, we begin our empirical analysis of the removal of step-up, guided by the framework and heterogeneity discussed above. Our reduced form estimates will combine short term and equilibrium responses, but we will return to a discussion of dynamics in Section 5.3.

5 Observed Responses to the Removal of Step-Up

In 2005, following thirteen years without substantial changes to capital gains and inheritance tax rates, Norway announced that step-up would be removed for inherited capital gains from stock (including private business interests and equity-based mutual funds).[35] Stock gifted or bequested after January 1, 2006, would be subject to capital gains tax on the original owner's gains when the heir next realized the asset.

In this section, we document responses to the 2005 announcement of the removal of step-

[35] See Appendix Table A2 for detail on minor modifications to the inheritance tax brackets during this time.

up for stock.[36] Aggregate capital gains realizations more than tripled between 2004 and 2005 (Appendix Figure A2), but year-to-year variation in aggregate realizations is frequently large and causal attribution requires a research design. We first estimate average treatment effects on the treated using a difference-in-differences design that exploits the model-implied age gradient in exposure (Section 5.1). To probe the robustness of the medium-run estimates from this design, which may be confounded from 2009 onward by changes to inheritance tax rates, we re-estimate effects using a second design that exploits heterogeneity in pre-2005 stock-portfolio appreciation (Section 5.2). The two designs produce similar aggregate results through the end of our data in 2018. We then use the model to interpret the dynamics implied by our estimates and document the anticipatory inheritance responses that shape them (Section 5.3). Finally (in Section 6.2), we show that although the realization response to the removal of step-up is large relative to typical revenue from the capital gains tax, the change is small relative to the stock of accumulated unrealized gains, which continue to grow throughout our post-period.

5.1 Age-Based Design

Our primary empirical strategy builds on two intuitions. Firstly (following Remark 1 from our model) we expect realization responses to the removal of step-up to be increasing in age, with younger age groups being effectively "untreated" by the removal of step-up. Secondly, we think the removal of step-up is relatively unusual in the way it affects the age-realizations gradient: we expect that normal year-to-year macroeconomic fluctuations would affect the realization behavior of individuals of different ages more proportionally. These intuitions lead us to a continuous differences-in-differences design (described formally below) in which treatment exposure is increasing in current age.

Our primary specifications collapse the continuous age gradient in exposure to a binary difference-in-difference design comparing groups of older and younger individuals. This bi-

[36] Although step-up was eventually removed for non-stock asset classes in 2014 (see Appendix F), evaluation of that removal is confounded by the 2014 removal of the inheritance tax, which had a simultaneous but opposite effect on the relative price of bequests versus realizations. There was also a capital gains tax cut in 2014. We therefore choose to focus our analysis only on the earlier reform, which removed step-up for stock while holding inheritance and capital gains tax rates fixed.

narization allows us to identify summary average treatment effects for the group of older individuals (Callaway et al., 2025) and to cleanly map those effects into an overall impact on capital gains tax revenue in Norway. However, we also examine dose-response relationships (i.e., heterogeneous treatment effects by age) and find them to be consistent with our theoretical predictions, which we interpret as bolstering the case for our causal claims.[37]

5.1.1 Identifying Assumptions

To identify our treatment effects, we will assume that if step-up had not been removed, different age groups' average realizations would experience the same percent changes over time in response to macroeconomic shocks and other time effects. Let A_{it} denote a normalized age variable with support $\mathcal{A}$, normalizing $A_{it} = 0$ as the control group of "young people" who are unaffected by treatment at time t. Our assumptions condition on X_i, a vector of time-invariant characteristics with support $\mathcal{X}$. (In practice, X_i denotes the individual's position in the age-specific wealth distribution and an indicator for private business ownership). We assume:

Assumption 1 (Parallel Trends in Percent Changes)**.** *The average realizations of individuals with age a would have, in the absence of treatment, evolved proportionally to the average realizations of the untreated younger age group ($A_{it} = 0$).*[38]

$$\frac{E[Y^0_{i,t}|A_{it}=a,X_i] - E[Y^0_{i,t_0}|A_{it}=a,X_i]}{E[Y^0_{i,t_0}|A_{it}=a,X_i]} = \frac{E[Y^0_{i,t}|A_{it}=0,X_i] - E[Y^0_{i,t_0}|A_{it}=0,X_i]}{E[Y^0_{i,t_0}|A_{it}=0,X_i]} \quad \forall t,\ a \in \mathcal{A},\ x \in \mathcal{X}$$

where $Y^0_{i,t}$ is individual i's potential outcome (realized capital gains) in year t in the absence of treatment and $t_0 < 2005$ is a fixed pre-treatment base year. This parallel trends assumption is equivalent to assuming that, in the absence of the policy change, yearly shocks may move

[37] We do not interpret differences between dose groups as an estimate of the average causal response to marginal exposure, which would require additional selection restrictions to identify (Callaway et al., 2025).

[38] An assumption of parallel trends in percent changes is commonly made by assuming parallel trends in a logged variable — but zeros in our outcome variable make a log transformation particularly inappropriate to our setting (Chen and Roth, 2024). Furthermore, estimating a diff-in-diff using OLS on a log transformation of the dependent variable only approximates the proportional difference in growth rates between treated and untreated groups. Such approximations ignore Jensen's inequality and thus may deliver biased estimates of the true multiplicative effect (Ciani and Fisher, 2019). Our estimator avoids this bias and can be estimated with Poisson Pseudo Maximum Likelihood (PPML) regression. PPML consistently estimates the multiplicative effect as long as the mean function is correctly specified as multiplicative (Wooldridge, 2023).

the age-log(realizations) profile up and down, but the curvature of the profile (i.e., the relative realization levels of different ages) will be stable over time. We investigate how the shape of the observed age-log(realizations) profile changed over time in Figure 2 Panel (b), comparing each age's average realizations to the average realizations of 30-to-40 year-olds in the same year. In the pre-reform period (before 2005), the shape of this profile appears to be roughly stable across years. Averaging across pre-period years, realizations peaked around age 54 at a level approximately 94% higher than those of 30-to-40 year-olds, and then decreased for older ages, so that average realizations at age 70 were approximately 5% below the realizations of 30-to-40 year-olds. This inverted-U relationship is consistent with the model-implied profile from our model in Figure 2 Panel (a). Figure 2 Panel (b) also provides initial evidence that this relationship changes in the initial years after the announcement of step-up's removal in 2005: realizations peaked at age 60 at a level approximately 160% higher than those of 30-to-40 year-olds and declined slowly, so that realizations by 70-year-olds were 119% higher than those of 30-to-40 year-olds in 2005-2008.

We also require a "no anticipation" assumption for the years we consider to be our pre-period:

Assumption 2 (No Anticipation Before 2005)**.**

$$E[Y_{i,t}^0 | A_{it} = a, X_i = x] = E[Y_{i,t}^a | A_{it} = a, X_i = x] \quad \forall t < 2005,\ a \in \mathcal{A},\ x \in \mathcal{X}$$

where $Y_{i,t}^a$ is individual i's potential outcome (realized capital gains) in year t if he experienced an exposure level like that of age a. Conditioning on X_i is only meaningful if untreated individuals are observed at every covariate value attained by the treated, so we additionally require:

Assumption 3 (Overlap)**.**

$$P(A_{it} = 0 \mid X_i = x) > 0 \quad \forall x \in \mathcal{X}$$

Under Assumptions 1 – 3, we can employ a difference-in-difference estimator to identify a covariate-specific average treatment effect $ATT_t(a,x) = E[Y_{it}^a - Y_{it}^0 | A_{it} = a, X_i = x]$, and hence its aggregation over the covariate distribution of the treated, $ATT_t(a) =$

$E\big[ATT_t(a, X_i) \mid A_{it} = a\big]$, for each age a.[39] We can also identify a binary summary effect for all "old" versus "young" individuals: $ATT_t = E[ATT_t(a)|A_{it} > 0]$.[40] We are primarily interested in the ATT_t in levels, because it translates directly into the aggregate increase in capital gains tax revenue. However, the same assumptions allow us to identify the effect of the removal of step-up in terms of percent changes from the counterfactual.

5.1.2 Control & Treatment Group Definition

Although our model implies that younger age groups will have less exposure, the model does not tell us what age cutoff we should use to form the control group of plausibly-unexposed younger individuals. To interpret our treatment effects as exact estimates, we need to believe that the younger ages we pick to form our control group are truly untreated by the removal of step-up. If the younger age group is partially treated and increases realizations in response to the removal of step up, the treatment effects we estimate for the older groups will understate the true treatment effects, since the comparison will net out any realization increase from the younger group (Callaway et al., 2025). In practice, we set up a difference-in-difference empirical strategy using younger 30-49 year olds as our control group and test our estimates' robustness to the choice of other age cutoffs in Appendix Figure A4. We accept the fact that our estimates may be lower bounds if 30-49 year-olds are partially treated. We compare our control group to "treated" individuals of older ages. Our summary (binary) regressions compare all individuals below 50 years of age to all older individuals. To examine dose-responses, we bin individuals into age groups. The choice of bin is arbitrary, so we present results using both decade age bins (e.g., 50-59, 60-69, etc) and deciles of the observed age distribution. Under Assumptions 1-3, we identify an average treatment effect for each older age group.

[39] Our $ATT(a)$ measures the average difference in potential outcomes between receiving dose a and dose 0 for the age group that was actually treated with dose a.

[40] Callaway et al. (2025) refer to this summary parameter (in the absence of additional restrictions on heterogeneous treatment effects) as ATT^{loc}. It is "local" in the sense that it is built from each age group's effect at the dose that age group actually received, $ATT(a|a)$, rather than from the effect that dose a would have had on the entire treated population, $ATT(a)$. Given our focus on the policy impact of the removal of step-up, this parameter is the right one for estimating the causal increase in capital gains revenue from Norway's actual policy change.

5.1.3 Estimation

Our primary outcome is net realized capital gains as reported on the individual's annual tax return, which includes realized capital gains from stock and from all other taxable asset categories.[41] We first estimate a binarized regression that groups all individuals older than 49 ($A_{it} > 0$) individuals together:

$$Y_{it} = exp\Bigg(\beta_0 + (\mathbb{1}[A_{it} > 0] \times Post_t)\beta_{1,t} + \mathbb{1}[A_{it} > 0]\beta_2 + Post_t\beta_{3,t} + \dot{X}_i'\beta_4\Bigg)\epsilon_{it} \tag{3}$$

where X_i are time-invariant control variables — a saturated set of indicators for the individual's position in the age-specific wealth distribution and for private business (unlisted stock) ownership — entering the specification in demeaned form, $\dot{X}_i \equiv X_i - E[X_i \mid A_{it} > 0]$; $Post_t$ is an indicator that equals one if the year is greater than or equal to 2005 (the announcement year); and $E[\epsilon_{it}|A_{it}, Post_t, X_i] = 1$.[42] We also estimate year-by-year event study estimates by replacing the $Post_t$ variable in the equation above with an indicator variable for each specific year. The exponentiated coefficient $\exp(\beta_1) - 1$ gives the percent change in realizations attributable to the reform, and we can construct the summary average treatment effect in levels as

$$ATT_t = exp(\beta_0 + \beta_{1,t} + \beta_2 + \beta_{3,t}) - exp(\beta_0 + \beta_2 + \beta_{3,t}) \tag{4}$$

To examine how responses vary by age, we run the following regression:

$$Y_{it} = exp\Bigg(\beta_0 + \sum_{a \in \mathcal{A}^+} (\mathbb{1}[A_{it} = a] \times Post_t)\beta_{1,ta} + \sum_{a \in \mathcal{A}^+} \mathbb{1}[A_{it} = a]\beta_{2,a} + Post_t\beta_{3,t} + \dot{X}_i'\beta_4\Bigg)\epsilon_{it} \tag{5}$$

with $\dot{X}_i$ defined as above and $E[\epsilon_{it}|A_{it}, Post_t, X_i] = 1$. We recover the percent change for each age group $A_{it} = a$ and corresponding $ATT_t(a)$from this regression. All regressions are estimated with Poisson Quasi-Maximum Likelihood (PPML) and restrict the sample to individuals we observe ever owning stock in the transactions data.

[41] The tax data nets out gains and losses across all realizations within an individual and year. We set net losses to zero in our regressions.

[42] We demean our control variables using the treatment group mean, so that $\dot{X}_i = 0$ for an individual at the treated group's average covariate values.

5.1.4 Threats to Identification in the Age-Based Design

Our age design identifies the increase in realized capital gains from the removal of step-up if yearly average realizations for younger and older age groups would have moved proportionally (i.e., experienced the same percent changes over time) but for the removal of step-up. In practice, however, we are concerned with several confounding policy changes in our post-period that may have differential age effects. Norway cut inheritance tax rates in half in 2009 and then fully repealed the inheritance tax in 2014. Older individuals are differentially exposed to these cuts, and the cuts push against the realization response to the removal of step-up; we therefore expect this design to deliver downward-biased estimates of the medium-run effect from 2009 onward. The alternative design in Section 5.2 addresses these issues by using cross-sectional exposure variation from portfolio composition conditional on age and wealth rather than age itself. In practice, both designs show similar results across the entire post-period.

5.1.5 Short-Term Results (2005-2008)

Table 2 Panel (a) presents estimated short-run responses from difference-in-difference regressions. We estimate large effects: our summary measure implies that individuals aged 50 and above increased realized capital gains by 36%, or \$1,068 a year on average. These effects are large compared to existing estimates of realization responses to capital gains tax cuts (see Appendix E). Effects are initially increasing in age, from $\beta_1 = 0.187$ for ages 50-59 to (a percent change of 21%) $\beta_{1,a} = 0.572$ for ages 70-79 (a percent change of 77%). The estimated effect for individuals aged 80-90 is statistically indistinguishable from that of the 70-79 age group. These coefficients translate into level increases in taxable capital gains realizations ($ATT(a)$) ranging from \$740 for individuals aged 50-59 to $\$1,592$ for individuals aged 70-79.[43] Figure 4 Panel (a) depicts age-specific difference-in-difference coefficients for finer age groups (deciles of the observed over-50 age distribution). Effects are zero around age 50 but increase to a peak of $\beta_1 = 0.6$ (82%) around age 70, remaining fairly stable for the rest of the age distribution. All results in this section control for the individual's position in

[43]Note that, despite similar percent changes, the level response is larger for the 70-79 age group than the 80-89 group due to higher baseline realizations.

the age-specific wealth distribution and for private business ownership and are run on the population aged 30 and above who ever own stock.

Table 3 explores what type of capital gains drive the increased realizations following the removal of step-up. For a subset of our data years, we are able to decompose our outcome variable of total capital gains into capital gains coming from real estate, from unlisted stock and from a residual category that is primarily listed stock and mutual funds.[44] Capital gains realizations from taxable real estate show small, mostly statistically-insignificant negative effects, which serves as a useful placebo: real estate was not affected by the initial removal of step-up. Capital gains realizations from unlisted stock reach conventional levels of statistical significance only for the oldest age group, where they account for roughly 25% of the total increase. For younger age categories, it is less clear whether there was a response in unlisted stock: results are small, statistically insignificant and sometimes negative. The response to the removal of step-up appears to be concentrated in the residual category dominated by listed stock and mutual funds, suggesting that the realization response came primarily from liquid asset sales rather than sales of private business interests. This is despite the fact that the majority of unrealized capital gains are not in liquid assets (Table 1).

5.1.6 Medium-Run Results (2009-2018)

Figure 2 Panels (b)-(d) presents event study results from the age-based design which extends through 2018. Effects are clearly strongest for older ages. pre-trends appear flat, the realization response begins sharply in 2005, and the yearly coefficient estimates remain elevated through the end of our window with minimal decay toward the pre-period level.[45] The persistence of medium-run effects through 2018 despite expected effects from short-term rebalancing (see Section 5.3) and the downward bias we expect from the post-2008 inheritance tax cuts is striking. We return to this finding in Section 5.2, where the UCG-based design provides a complementary medium-run estimate, and in Section 5.3, where we use the model

[44] Our data on capital gains from specific asset classes does not fully cover the same years as our primary outcome measure of total capital gains. We explain these limitations in more detail in Section 1 and in the note to Table 2.

[45] Note that while the event study coefficients do not decrease over time, yearly ATT estimates experience some downwards movement (Figure 4 Panel (b)), due to changes to the level of the control group mean, especially during the Great Recession.

to interpret the dynamics over time.

5.2 UCG-Based Design

Individuals' exposure to the inheritance tax cuts in our post-period depends on their age and the size of their expected bequest. We are therefore unable to disentangle the longer-run effects of the removal of step-up from changing inheritance tax incentives in 2009 and 2014 in our age-based design. Remark 2 of our model (Section 4), however, points us to an alternative source of exposure variation to the removal of step-up: we expect individuals with the same age and wealth level but with different levels of unrealized capital gains in that wealth to be differentially exposed to the removal of step-up. These individuals would not be differentially exposed to inheritance tax changes, because inheritance taxes are imposed on terminal wealth without regard to the amount of UCG in that wealth.

To build on this insight, we would ideally compare the responses of individuals with different unrealized capital gains but the same age, wealth and purchase history. However, since UCG are mechanically determined by purchase prices and current values, there is no residual variation among people with the same current holdings and identical transaction histories. To disentangle exposure to the removal of step-up from exposure to inheritance tax changes, we condition on age and current wealth, accepting that the residual variation is necessarily variation in transaction histories – either through different initial investments in the market (later equalized by variation in returns) or different rebalancing behavior after the initial investment.

5.2.1 Identifying assumptions.

This design rests on the identification assumption that among individuals holding stock at the end of 2004, realization trends, but for the removal of step-up, would have had a stable gradient in gamma (the ratio of unrealized capital gains to wealth in 2004) over time. Formally, let G_i denote individual i's ratio of unrealized capital gains in stock to total stock value at the end of 2004, with support $\mathcal{G}$ and $G_i = 0$ for the control group.[46] We require

[46] We use baseline $G_i = \gamma_{i,2004}$ as a predetermined proxy for the model-relevant exposure at death, $U^{+}_{i\mathcal{T}}/W_{i\mathcal{T}}$, maintaining that, conditional on age and total wealth, expected exposure at death is increas-

analogues of Assumptions 1–3, with G_i replacing A_{it} and with X_i expanded to include age bins alongside the wealth and private-business indicators. In words, we assume that in the absence of the reform, average realizations of investors with $G_i = g > 0$ would have experienced the same percent changes as those of the $G_i = 0$ group (investors with zero or negative end-of-2004 appreciation) with the same age and wealth. Fixing G_i at its end-of-2004 value ensures that treatment intensity is determined pre-announcement and is not endogenous to the realization response itself.

5.2.2 Estimation.

We restrict the sample to individuals holding stock at the end of 2004. As in Section 5.1, the outcome is again net realized capital gains as reported on the individual's annual tax return. We estimate the PPML specifications of Equations [5] and [3] with G_i-bin indicators in place of age-bin indicators, and an event study specification interacting treatment with year. Our binarized treatment group pools all investors with $G_i > 0$. Because this design requires conditioning on age as well as wealth, the control vector $\dot{X}_i$ here contains saturated age-bin indicators in addition to the wealth-position and private-business indicators of Section 5.1, and is demeaned using the mean of the $G_i > 0$ treatment group.

5.2.3 Threats to Identification in the UCG-Based Design

This design compares individuals within the same age and wealth group who have different unrealized capital gains ratios (gamma) at the end of 2004. Necessarily, high-gamma individuals will have either made smaller initial investments and earned more on those investments or will have rebalanced less than their low-gamma counterparts. Group definition is therefore selected on a mixture of return luck, differences in initial investments and past non-realization.[47] Persistent traits between these groups of selected individuals (such as a persistently higher willingness to realize) will be netted out by our first differences. However,

ing in baseline within-stock appreciation. Since $U^+/W = \omega, \gamma^+$ scales with both within-stock appreciation (γ^+) and the portfolio share of stock (ω), this requires that baseline appreciation persists and that higher-G_i investors do not hold systematically smaller equity shares over the remaining lifecycle.

[47] Alternative choices of conditioning sets (such as an attempt to isolate the variation coming from return luck by conditioning on initial investments) necessarily let wealth depend mechanically on treatment intensity, reopening the differential exposure to the inheritance tax cuts that this design exists to close.

correlation between the return histories that generate high and low gamma and those individuals' responses to time variation remains a threat to identification in this setting. Event study pretrends let us detect, but not rule out, differential trends from such selection.

5.2.4 Results from the UCG-based Design

Table 4 Panel (a) presents the estimated short-run (2005–2008) responses for the UCG-based design. Column (1) estimates a difference-in-difference regression on all individuals who owned stock at the end of 2004 with a binary treatment variable equal to one if the individuals' end-of-2004 stock portfolio contains positive unrealized capital gains ($\gamma > 0$).[48] Individuals with positive pre-2005 appreciation increased their realized capital gains by 54% ($1,791/year on average) relative to the control group. Figure A3 plots event study coefficients separately by quartile of G_i: the bottom two quartiles show no statistically-significant response, the third quartile shows a moderate response, and the top quartile drives the bulk of the aggregate effect. The pattern is monotone in G_i, consistent with Remark 2. Columns (2) and (3) of Panel (a) re-estimate the same specification restricting the sample to individuals who were less than or, respectively, at least 50 years old in 2005 (birth cohorts 1955 and earlier). Restricting to individuals who were at least 50 in 2005 serves two purposes. Substantively, it isolates the population for which the removal of stepup is most immediate. Practically, it also places the UCG design on (approximately) the same population as the age design, allowing the two designs' implied aggregate effects to be compared directly.[49] Panels (b) and (c) present event study estimates corresponding to the difference-in-difference regressions of Columns (2) and (3). As in the age-based design, pre-trends appear flat, the response begins sharply in 2005, and effects remain elevated through the end of our window with no decay toward the pre-period level. In the UCG design, like the age design, the medium-run response does not decay relative to the short-run response. We take up the dynamics of the results again in Section 5.3.

[48] Appendix Figure A4 shows robustness to re-defining the treatment and control groups with alternative $\gamma > 0$ cutoffs.

[49] The populations are only approximately the same because the age design defines age groups based on contemporary ages, whereas column (2) of Table 4 subsets by age in 2005 (consistent with its treatment variable being fixed at end-of-year 2004). Over the 2005-2008 window, however, the two definitions nearly coincide: the year-t population aged 50+ differs from the 2005 cohort aged 50+ only through individuals aged 47-49 in 2005 who cross the threshold during the window.

5.2.5 Results Comparison with the Age Design

The UCG-based and age-based designs estimate different objects on different populations, so their coefficients are not directly comparable: the age design dilutes the response of the exposed few across the entire 50+ population, while the UCG design conditions on exposure. The comparable objects are implied aggregates. Over 2005-2008, the age design implies a total increase in realizations of \$434 million; the UCG design restricted to the same cohorts implies \$501 million. These need not coincide even in principle, because each design nets out its own comparison group's response: the age design subtracts any response of individuals under 50, some of whom may hold embedded gains and face the reform's incentives through gifts and expected future bequests, while the UCG design subtracts only the response of zero- and negative-gain holders, for whom the reform's incentive is smallest. Consistent with this, the unrestricted UCG design (Column 1) implies a larger aggregate than the age design, with the gap accounted for by younger gain-holders' estimated response. On the other hand, given the event study pattern in Panel (b), we are not very willing to causally attribute an additional \$180 million dollars in realizations to the Age < 50 population. We therefore read the age design's aggregate as a lower bound on the total response, take it as our headline revenue figure on the grounds of its cleaner identifying assumption, and read the UCG design as establishing where the response is concentrated (in embedded gains) and that it extends below the age threshold, which is precisely why the headline figure is a lower bound.

5.3 Interpreting the Dynamics: Anticipation and the Long Run

The two designs of Sections 5.1 and 5.2 produce a common, and at first surprising, finding: the medium-run realization response is not smaller than the short-run response. A textbook intuition about the removal of step-up would predict a large transitional spike in 2005, as investors rebalance away from positions that were optimal under step-up, followed by a gradual decay toward a lower equilibrium level of realizations. Neither design shows this pattern: in both, the response is sharp at onset and persistent through 2018.

The model simulations reconcile the finding once anticipation is taken seriously. Under an unanticipated removal of step-up (no early transfers), the model does produce a large transition-year spike as investors rebalance to the lower equity holdings appropriate to a carryover regime. But Norway's removal was anticipated: Norwegians could use the year-long lag between announcement and implementation to make *inter vivos* transfers that still benefited from step-up. Extending the model to allow anticipatory transfers (as in Appendix B) significantly dampens the 2005 spike.

5.3.1 Empirical evidence of anticipation.

If anticipatory transfers in 2005 absorbed gains that would otherwise have been realized (and would have otherwise caused a short-term spike), we should see an outlier volume of intergenerational stock transfers in the months between the December 2004 announcement and the January 2006 implementation. Figure 3 Panel (a) documents exactly this: inheritance transfers containing unrealized capital gains in stock spiked in the second half of 2005, far above any level observed before or after. Contemporary press accounts urged Norwegians — particularly private business owners — to consider early inheritance transfers (Amundsen, 2005; Vigdal, 2005), and the data show the advice was widely taken up.

5.3.1.1 Implications for other institutional settings. The anticipation channel also disciplines what we should expect from removals of step-up in countries whose tax systems grant step-up only to bequests at death and not to *inter vivos* gifts (including the United States and the United Kingdom). [50] A removal of step-up in such a setting precludes the anticipation channel: the transition path would resemble the unanticipated simulation, with a larger short-run spike and faster convergence to the same long-run level. The implication for the external validity of our results is that the long-run realization response to the removal of step-up may be roughly transportable across institutional contexts, but the shape of the transition path is not.

[50] The anticipatory responses were only possible because of the lag between the announcement and implementation of step-up's removal and the fact that Norway allowed step-up for *inter vivos* gifts.

5.3.2 Relation to the 2006 Tax Reform

The repeal of step-up was enacted as part of Norway's broader 2006 tax reform, which also changed the taxation of shareholder and labor income. Other components of the reform could in principle confound our estimates, but several features of our empirical strategy mitigate this concern. First, our estimates do not rely on the aggregate change in realizations around the reform. Year fixed effects absorb shocks and reform responses common across the treatment and comparison groups. Second, the two designs exploit distinct dimensions of exposure. Thus, to account for the joint evidence across both designs, alternative reform channels would need to generate both the age gradient and the portfolio-appreciation gradient that we estimate. Third, the asset decomposition provides additional reassurance, as the realization response is concentrated in listed stock and mutual funds rather than private businesses.

6 Aggregate Implications

The removal of step-up has two contrasting aggregate consequences: a large response in capital gains realization flows, but a small change to the stock of unrealized capital gains. Sections 6.1 and 6.2 document this asymmetry. Section 6.3 translates these measures of stock and flow into welfare implications in a marginal value of public funds framework. Section 6.4 discusses alternative policies and the external validity of our findings.

6.1 Aggregate Effects on Capital Gains & Inheritance Tax Revenue

We convert our realization results into estimates of total tax changes by calculating average treatment effects on the treated (following Equation 4, included in Tables 2 and 4) and multiplying that average by the total number of treated individuals. Our baseline results (Table 2) imply a short-run (2005-2008) increase in realized capital gains of $434 million dollars ($1,068 ATT $\times$ 406,496 Treated Individuals). For the subset of individuals aged 50+, the UCG design estimates in Table 4 imply a similar rise in aggregate tax revenue: $501 million. This implies that 15% of the (notably high) level of taxable capital gains in 2005 ($2.96 billion) was attributable to the removal of step-up and that the removal of step-up can be said to have increased capital gains tax revenue 17% above counterfactual in 2005. Figure 4 shows that our event study ATT estimates for both designs are fairly stable over time.

6.2 Aggregate Unrealized Capital Gains in the Post-Period

Above, we documented an increase in realizations of highly-appreciated stock by the original owners. A natural question is whether this increase translates into a meaningful reduction in the aggregate stock of unrealized capital gains. Figure 5 shows that it does not: the stock of unrealized capital gains in stock and taxable real estate continues to grow throughout our post-period, despite the higher level of taxable realizations.

This pattern, which might initially seem at odds with our realization results, is in fact consistent with the relative magnitudes of the stock and the flow. Under the accounting identity previously articulated by Gravelle (1991), the year-to-year change in unrealized capital gains under step-up is:

$$UCG_{t+1} = UCG_t + AccCG_t - RCG_t - ICG_t \tag{6}$$

where $AccCG_t$ is capital gains accrued during year t from appreciation of existing holdings and acquisition of new positions, RCG_t is taxable realizations, and ICG_t is capital gains transferred in inheritance.

Equation 6 makes clear that the year-to-year change in unrealized capital gains is the difference between a single inflow (net accrual) and two outflows (realization and inheritance). In 2004, both outflows were small relative to the stock: $910 million of taxable realizations and $550 million of inherited capital gains, against a stock of $64.85 billion in stock and taxable real estate (Table 1). Our estimated realization response ($434 million per year, or 48% of baseline RCG) adds an additional outflow equivalent to 0.7% of the 2004 stock per year.

To benchmark the inflow side of Equation 6, we apply a deliberately conservative real return of 5% per year to the \$109 billion of underlying asset value in 2004.[51] This generates an estimated $AccCG_t \approx \$5.44$ billion per year — more than three times the combined annual outflow from realization and inheritance. The implication is that, even after accounting for our estimated realization response, the stock of unrealized capital gains would be expected to grow by roughly \$3.55 billion (5.5%) per year through this period.[52]

A useful thought experiment is to consider what would have happened if the removal of step-up had *doubled* aggregate taxable realizations, rather than producing the 15% increase we estimate. The additional \$910 million of annual outflow would amount to 1.4% of the 2004 stock per year. Even under this extreme counterfactual, the implied net change in UCG_t would remain positive: at a 5% accrual rate, the stock of unrealized capital gains would still grow by roughly \$3.07 billion (4.7%) per year.

This stock-versus-flow comparison reconciles the two facts. Removing step-up generated a large response in capital gains tax revenue because taxable realizations are small relative to the underlying stock of accumulated, unrealized appreciation. The same arithmetic implies that even a much larger behavioral response would only modestly change the level of the UCG stock — the policy reform pulls forward gains that would otherwise have remained in the stock, but at a pace that is small relative to ongoing accrual.

Under carryover, the inheritance term ceases to be an outflow: transferred gains are no longer forgiven, and they remain in the stock in the hands of the heir. The reform therefore removes \$550 million per year of outflow from the stock while adding only \$434 million per year of induced realizations. Mechanically, removing step-up *raises* the growth rate of the aggregate stock of unrealized capital gains.

The realization response captures revenue from gains that would otherwise have been inherited tax-free, but it does not meaningfully shrink the stock of accumulated unrealized appreciation in the economy. For policymakers whose concern is the size of that stock — rather than the revenue raised on its eventual outflow — removing step-up is at best a

[51] We view 5% as a conservative benchmark for two reasons. First, it is well below realized Norwegian listed-equity returns over the period we study: the Oslo Børs benchmark index more than doubled in real terms between 2004 and 2007. Second, holding $AccCG_t$ fixed at 5% applies the appreciation rate only to the 2004 asset value, ignoring growth in the asset base itself over time.

[52] Our calculated growth rate for UCG exceeds the assumed 5 percent accrual rate because appreciation accrues on the full \$109 billion of underlying asset value, while unrealized gains are only \$64.85 billion of that total.

partial instrument: the bulk of the stock is sustained by the realization principle itself, not by step-up.

These aggregate flows pin down the ingredients for the welfare analysis that follows: the next section combines the annual realization response with the annual flow of capital gains passing through inheritance, and the inheritance tax rate τ_I, to evaluate the welfare consequences of step-up.

6.3 Marginal Value of Public Funds

We consider the implications of our results for social welfare in a marginal value of public funds (MVPF) framework (Wildasin, 1984; Mayshar, 1990; Hendren, 2016): $MVPF = \frac{1}{1+FE}$ where the fiscal externality FE is the ratio of government revenue effects from behavioral responses relative to mechanical government expenditure on the program. Using the notation of our model (Section 4), we derive an upper bound for the MVPF of a marginal increase in carryover policy θ as: [53]

$$MVPF_{d\theta} = \frac{1}{1 + FE_{d\theta}}; \qquad FE_{d\theta}(\theta) = (1 - \theta - \tau_I)\frac{dR_{T-1}/d\theta}{U_T(\theta)} \tag{7}$$

Each marginal dollar of induced realization R raises capital gains revenue by τ_c today, forgoes the $\theta\tau_c$ that the heir would have paid on those gains, and forgoes $\tau_I\tau_c$ of inheritance tax because the capital gains tax paid shrinks the taxable estate. The denominator $U_T(\theta)$, the annual flow of unrealized gains transferred at death, is the mechanical base of the tax expenditure.[54]

However, since we evaluate a discrete policy change from $\theta = 0$ (step-up) to $\theta = 1$

[53] Appendix Section C walks through the derivation of Equation 7. Relative to our dynamic model in Section 4, the derivation restricts attention to an individual with positive terminal gains who considers whether to sell in his final period – restrictions that define the margin exposed to Norway's gains-only carryover – and adds two behavioral simplifications, each of which raises the MVPF. We assume heirs realize inherited gains immediately, though in practice they defer — deferral shrinks the present value of both the tax expenditure and the heir-side revenue offset, raising FE. We also assume the older generation's consumption does not respond to the reform, though any positive response adds recaptured inheritance tax revenue. Equation 7 is therefore an upper bound on the marginal MVPF within the model.

[54] Because Norway's carryover applied only to gains (losses retained a basis reset), all U_T objects here are positive-gains constructs, $U_T^+ \equiv \max\{U_T, 0\}$ in the notation of Section 4; we suppress the superscript. The empirical counterparts are constructed from positive inherited gains only.

(carryover), we follow Kleven (2021) and approximate the upper-bound $MVPF$ for the full removal of step-up by assuming linearity between endpoints for both willingness-to-pay and mechanical revenue:

$$MVPF_{0\to1} = \frac{1}{1+FE_{0\to1}}; \qquad FE_{0\to1} \approx \big(\frac{1}{2} - \tau_I\big)\frac{\Delta R_{T-1}}{\frac{1}{2}\big(U_T(0)+U_T(1)\big)} \tag{8}$$

We measure the elements of this equation as steady-state annual flows in 2018 dollars. For ΔR_{T-1}, we take the age-specific medium-run (2009–2013) realizations responses from Figure 4, which in aggregate yield \$447 million per year of additional taxable realizations before death. For $U_T(1)$, we measure the inherited capital gains in the post-period directly: capital gains in inherited stock averaged \$722 million per year over 2009–2013. For $U_T(0)$, we scale the pre-reform (2003–2004) flow of capital gains in inherited stock by the growth of inherited stock values between the two windows, yielding \$654 million per year.With $\tau_I = 0.10$ (the top inheritance tax rate in force during the 2009–2013 measurement window), these numbers deliver $FE = 0.26$ and an MVPF of 0.79 (Appendix Table A6 collects all inputs, units, and sources).[55] The result is stable across measurement choices: using the pre-2009 top inheritance tax rate ($\tau_I = 0.20$) gives 0.84; an alternative package built entirely from the short-run era (2005–2008 responses, 2006–2008 inheritance flows, and the 20% rate then in force) gives 0.89; and varying the growth adjustment between no growth and a doubling moves the estimate by less than 0.04 in either direction. Relaxing the simplifying assumptions for the derivation lowers the MVPF — under plausible heir deferral horizons, to between roughly 0.5 and 0.7 (Appendix C). The conclusion that survives every variant is that the MVPF of step-up is below one: step-up not only redistributes toward inheritors of appreciated assets, but actively amplifies the distortion from the capital gains tax. A

[55] Equation 8 bounds the MVPF from above within the model; a separate question is whether our empirical mapping preserves the bound. The formula prices a realization in the final period before transfer, but our measured ΔR_{T-1} aggregates responses by investors aged 50 and above, many of whom will live for decades. Applying the terminal-period coefficient to these responses treats each induced dollar as reducing eventually-inherited gains one-for-one, with the heir's forgone tax counted at face value. Both conventions continue to reinforce interpretation of our MVPF estimate as an upper bound. Counting the heir-side offset at face value overstates its present value. And while a dollar realized and reinvested decades before death can reduce eventually-inherited gains by *more* than a dollar (the capital gains tax paid stops compounding), that excess is exactly the future appreciation whose face-value taxation the first convention overstates: pricing the heir-side flows consistently, the product of the two never exceeds the one-for-one benchmark (Appendix C.1.2).

corollary is that the revenue-maximizing capital gains rate is lower under step-up than under carryover, since step-up raises the elasticity of taxable realizations.

An MVPF below one is unusual for a tax expenditure, and it stands in sharp contrast to MVPFs for tax cuts, which typically exceed one because they reduce distortions (Hendren and Sprung-Keyser, 2020). Step-up does the opposite: it provides a tax benefit while amplifying the distortion from the capital gains tax. This comparison points to a budget-neutral reform. Repealing step-up and returning the revenue as a reduction in the capital gains tax rate would benefit a similar (somewhat broader and somewhat less wealthy) population of capital gains owners, while reducing rather than amplifying distortions.[56] Given the concentration of step-up benefits in the top of the wealth distribution (Section 2) and the existence of the rate cut alternative, step-up is difficult to justify on either equity or efficiency grounds.

6.4 External Validity

Although our data and policy changes take place in Norway, we believe our findings are relevant to ongoing policy debates about the taxation of inherited capital gains in other countries. We find a large increase in taxable realizations when step-up is removed, which persists throughout our post-period, even after the inheritance tax is fully repealed in Norway. Because Norway granted step-up to *inter-vivos* transfers, the removal of step-up in Norway featured a year-long period during which owners of capital gains could make early inheritance transfers that would still benefit from step-up. The availability of this anticipatory response means that we expect smaller short-run realization responses in Norway, relative to what we would expect for the removal of step-up in other countries which grant step-up only for deathtime bequests. There is also arguably less incentive to defer capital gains in Norway, relative to countries such as the US and UK. This smaller incentive is due to the fact that inter-corporate dividends and capital gains are largely tax exempt, which allows Norwegians to build up holding companies in which they can reallocate their portfolios without triggering the capital gains tax.[57] Step-up was correspondingly less valuable in Nor-

[56] The revenue raised by repeal could finance a capital gains rate cut whose size depends on the elasticity of realizations within a carryover regime, which we expect to be smaller than the elasticity under step-up.

[57] Under Norway's exemption method (*fritaksmetoden*), corporate share gains are tax-exempt and qualifying dividends are largely exempt (only 3% is taxable).

way than in jurisdictions that tax intercorporate transactions, so a response of comparable magnitude in those settings would correspond to a larger underlying behavioral shift.

Furthermore, Norway levied an inheritance tax with low exemption thresholds during the stepped-up basis regime (see Appendix Table A2), meaning most people needed to pay inheritance tax to benefit from step-up. In contrast, the estate tax regimes currently in place in the US and UK exempt many inheritances from transfer tax while still granting them stepped-up basis. On the other hand, Norway's yearly wealth tax has been shown to encourage higher levels of savings (Ring, 2025).

As a final consideration, realization responses will differ across settings with different stocks of unrealized capital gains. Appendix Figure A5 compares our measure of pre-reform unrealized capital gains in Norway to the corresponding measure from the 2004 U.S. Survey of Consumer Finances, plotting average unrealized gains against average net wealth for groups defined by net worth percentile within each country. The two profiles lie close to one another throughout: Norwegian households hold comparable (slightly higher) unrealized gains than U.S. households of the same total wealth through the middle of the distribution, and the two converge among millionaires. This comparison suggests that Norway is not an outlier in unrealized capital gains that could be drawn into realization by the removal of step-up and, if anything, is a conservative benchmark for extrapolation to the United States. Because the gains exempted by step-up come overwhelmingly from the top of the wealth distribution (Section 2.4), the relevant region of Appendix Figure A5 is the upper tail, and there the U.S. wealth distribution extends past the range of the Norwegian distribution — with unrealized gains rising in proportion. The SCF moreover excludes the wealthiest U.S. households by design, so it understates the U.S. stock precisely where step-up is most valuable.[58] A U.S. reform would thus apply to a base of unrealized gains at least as large, and more concentrated at the top, than the one we study.

It is hard to know exactly how to adjust our estimates for these institutional factors, but the main point is that there is nothing obvious about our empirical setting that would lead us to expect smaller effects elsewhere. Our hope is that future research will be able to study

[58]SCF sampling excludes any household containing a member of the Forbes 400. Our Norwegian measure is administrative and has no analogous truncation, but the richest Norwegians are not as wealthy as the very richest Americans.

the removal of step-up in other countries, and therefore directly test the external validity of our findings. In the meantime, the conclusion of our policy comparison in Section 6.3 applies as long as our empirical results are *directionally* true. We have shown that step-up is a tax break that amplifies behavioral distortions. Reducing tax rates, in contrast, gives a tax break while simultaneously reducing tax-induced behavioral distortions. It is therefore hard to justify the continuation of step-up policies, even if the goal were to provide a tax break to certain capital-gains-rich individuals.

7 Conclusion

This paper provides the first quasi-experimental estimates of how the tax treatment of inherited capital gains affects realization and inheritance behavior, using novel data on unrealized capital gains in Norway and the 2005-2006 removal of step-up. We document that unrealized capital gains are a quantitatively large share of household wealth, especially at the top, and that the capital gains exempted under step-up represent a sizable fraction of the potential capital gains tax base. Two complementary research designs deliver estimates of a 17% increase in aggregate taxable realizations following the removal of step-up. The response is concentrated among the most-exposed ages and portfolios and persists undiminished through 2018.

The same response that is large in revenue terms is small relative to the stock it draws from. The additional flow of taxable realizations amounts to roughly 0.7% of the 2004 stock of unrealized capital gains per year, well below the rate at which new gains accrue. The stock of unrealized capital gains in Norway continued to grow throughout our post-period. Removing step-up captures revenue from gains that would otherwise have been inherited tax-free, but it does not meaningfully shrink the underlying stock of unrealized capital gains, which is sustained by increasing asset prices and the realization principle itself. That the stock persists is not, however, an argument for retaining step-up: because step-up raises the return to deferral, it amplifies rather than offsets the lock-in distortion created by the capital gains tax, and a budget-neutral repeal that returned the revenue as a lower capital gains tax rate would benefit a similar population of capital gains owners while reducing distortions rather than increasing them (Section 6.3).

We finish by reflecting on directions for future work. Our work to distinguish deferred capital gains from those that are truly "never-realized" is limited by the time horizon of our data: we would like to see more attempts to separate capital gains deferral from permanent escape. We would also like to see more work surrounding capital gains in private businesses. Unlisted stock holds most of the unrealized appreciation we measure and most of the appreciation we observe passing to heirs, but it supplies only a small and imprecisely estimated part of the realization response: gains from unlisted stock reach significance only in the oldest age group, contributing about a quarter of the total increase, and are indistinguishable from zero for younger owners. We cannot say whether this reflects responses to tax incentives, illiquidity, or motives for retaining family ownership that operate independently of tax. If the most appreciated assets in the economy are held in firms whose transfer is governed by who will run the business rather than by the tax cost of selling it, then reforms to the taxation of accrued gains will bind least exactly where the untaxed gains are largest. Understanding how unrealized capital gains and business succession interact in closely held firms strikes us as a valuable direction for future work.

Table 1: Unrealized, Realized, and Inherited Capital Gains in 2004 (billions of 2018 USD)

(a) Asset Value & Unrealized Capital Gains

Asset Type	Total Value	UCG	UCG/Value
Taxable Assets including Mutual Funds	118.95	75.05†	0.63†
Mutual Funds	10.20	-	-
Taxable Assets excluding Mutual Funds	108.75	64.85	0.60
Listed Shares	8.85	1.82	0.21
Unlisted Shares	89.80	58.90	0.66
Taxable Real Estate	10.10	4.13	0.41
Nontaxable Real Estate	374.00	151.00	0.40

(b) Realizations & Realized Capital Gains

Asset Type	Amount Realized	RCG	RCG/Amount
Taxable Assets excluding Mutual Funds	14.18	0.91	0.06
Listed Shares	12.30	0.31	0.03
Unlisted Shares	1.59	0.51	0.32
Taxable Real Estate	0.29	0.09	0.31
Nontaxable Real Estate	5.16	2.08	0.40

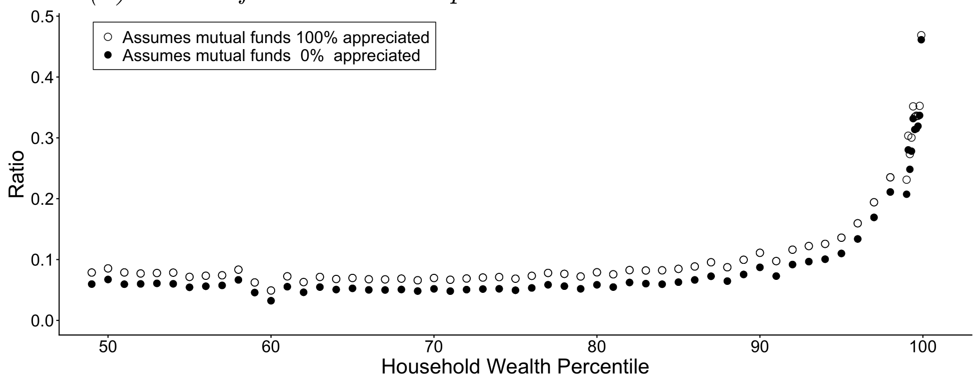
(c) Inheritances & Inherited Capital Gains

Asset Type	Amount Inherited	ICG	ICG/Amount
Taxable Assets excluding Mutual Funds	0.73	0.55	0.75
Listed Shares	0.05	0.03	0.69
Unlisted Shares	0.61	0.48	0.79
Taxable Real Estate	0.07	0.03	0.42
Nontaxable Real Estate	3.48	1.54	0.44

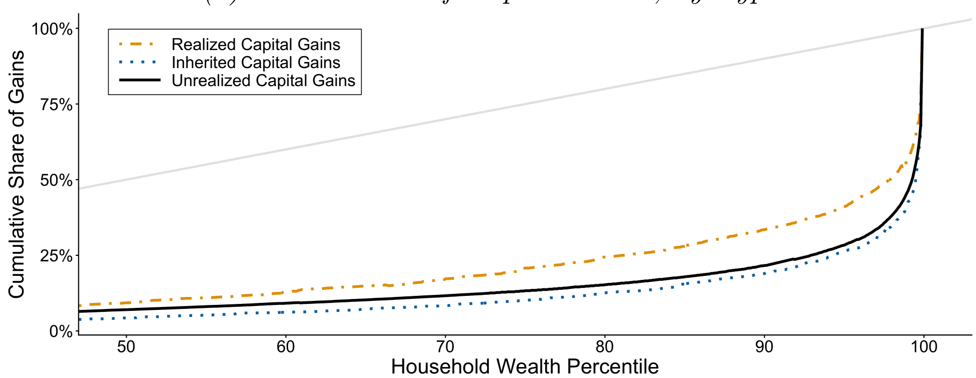
Note: This table presents aggregate numbers on the value of wealth and capital gains in certain asset categories in Norway at the end of 2004. Panel (a) presents the total amount of wealth ("Total Value") in each category, along with the amount of Unrealized Capital Gains ("UCG"). The amount of UCG when mutual funds are included assumes that mutual funds are 100% appreciated (estimates affected by this assumption are marked with a dagger †). Panel (b) displays the value of assets in these categories that are realized in 2004 ("Amount Realized"), along with the Realized Capital Gains ("RCG"). Panel (c) displays the value of assets in these categories that are inherited in 2004 ("Amount Inherited"), along with the Inherited Capital Gains ("ICG").

Figure 1: Capital Gains Across the Household Wealth Distribution, 2004

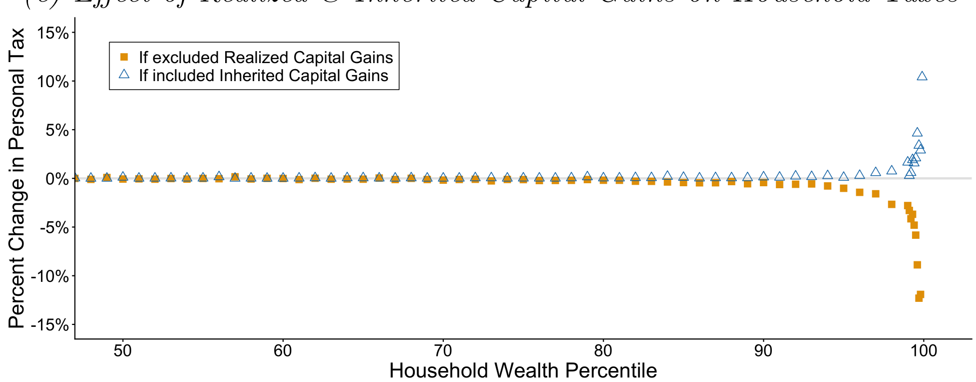
(a) Ratio of Unrealized Capital Gains to Household Wealth



(b) Distribution of Capital Gains, by Type



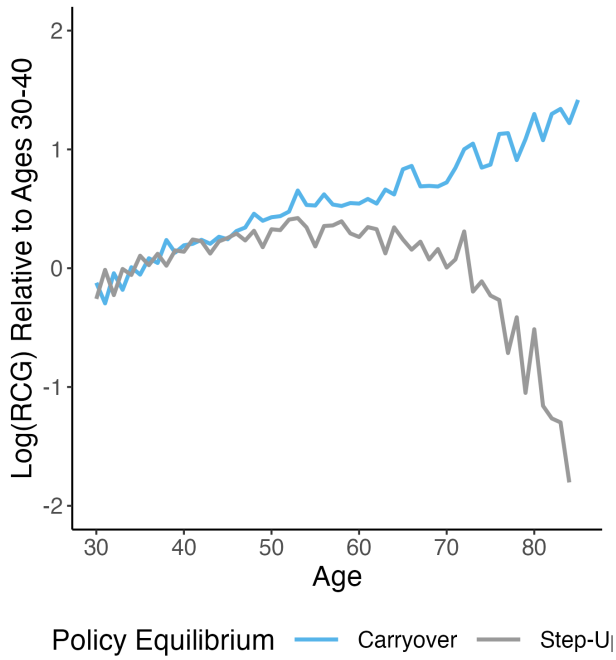
(c) Effect of Realized & Inherited Capital Gains on Household Taxes



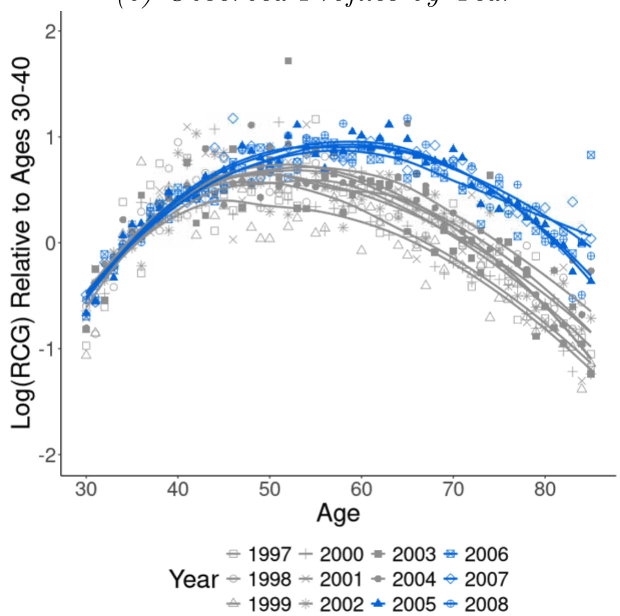
Note: This figure displays several ways in which capital gains vary across households, with households grouped on the x-axis by their total net wealth percentile in 2004. Panel (a) depicts average ratios of unrealized capital gains in assets subject to the capital gains tax (mutual funds, listed and unlisted stock and taxable real estate) to total household net worth. Because we do not observe unrealized capital gains in mutual funds, we bind our estimates for taxable unrealized capital gains under the alternative assumptions that mutual fund wealth is either 0% or 100% appreciated. Panel (b) compares the concentration of realized, inherited, and unrealized capital gains by plotting the proportion of gains held by the bottom x% of households, where household are ranked by their total net worth (a la Lorenz (1905)). The light gray line depicts perfect equality ($y = x$). Unrealized and inherited capital gains are significantly more concentrated at the top of the wealth distribution than the distribution of realized capital gains. Panel (c) quantifies the mechanical average effect that either excluding realized capital gains from the tax base or including inherited capital gains in the giver's tax base (at the regular capital gains rate) would have on households ranked by total net worth. Note that we are unable to observe inherited capital gains for mutual funds and that inherited capital gains are attributed to the gifting household.

Figure 2: Realized Capital Gains by Age, by Policy Regime

(a) Model-Implied Profiles by Policy Regime



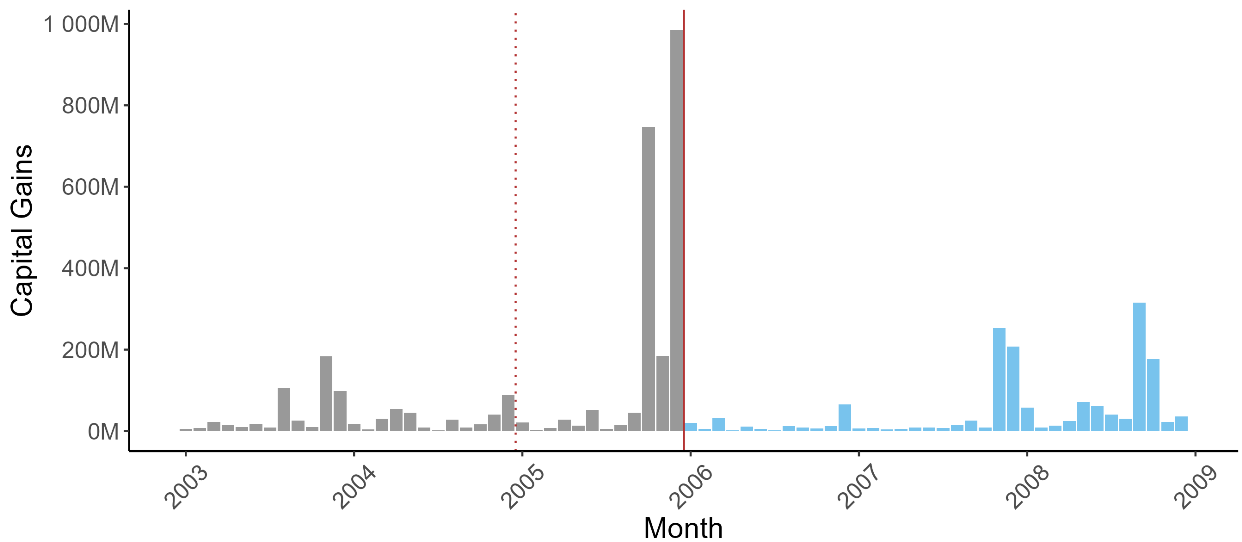
(b) Observed Profiles by Year



Note: This figure displays two graphs of the profile of log realized capital gains in age. Y-axis values are normalized to the average value for the 30-to-40 year-olds in the same year or simulation. Panel (a) plots profiles from our model simulation when either step-up or carryover is always in place (i.e., when the policy regime does not change during individuals' lifecycles). Parameterization details are described in Appendix B. Under step-up, individuals reduce capital gains realizations late in life. Under the carryover regime, individuals draw down their equity shares in retirement to consume their gains and shift their portfolios towards safer assets. Under the step-up regime, in contrast, individuals realize less capital gains at older ages. Panel (b) plots logs of the empirical average realizations by age and year observed in our data from 1994-2008. The points are year- and age-specific logged averages. The smoothed lines depict local nonlinear regressions (LOESS) of logged average realizations on age for each year or era. Years prior to 2005 (during step-up) are depicted in light grey and the years from 2005 to 2008 are depicted in dark blue. The shape of the profile is relatively stable from 1997 to 2004 (during step-up), but older individuals realized relatively more since 2005. We end this graph in 2008, before the 2009 decrease in inheritance tax rates.

Figure 3: Anticipatory Inheritance Transfers in Late 2005

(a) Capital Gains in Inherited Stock, by Month



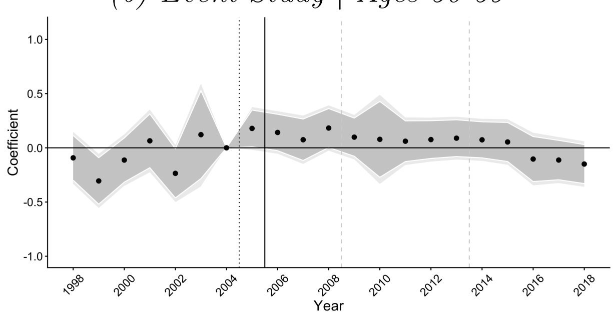
Note: This figure demonstrates the anticipatory transfers that occurred in late 2005, which were triggered by the announcement that step-up would be removed for stock on January 1, 2006. Transfers made prior to January 2006 would still be subject to stepped-up basis treatment. Panel (a) shows that the amounts of capital gains transferred in inheritance in the later months of 2005 were large outlier values compared to monthly values before or since. The dotted red line before 2005 indicates the announcement of step-up's impending removal for stock. The solid red line before 2006 indicates the implementation of step-up's removal for stock. Values are in 2018 USD.

Table 2: Realization Responses from Baseline (Age-based) Empirical Design

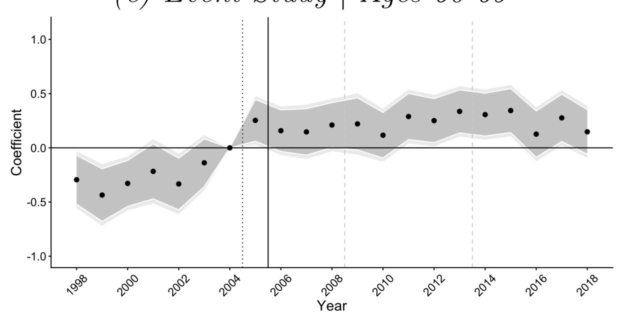
Panel (a): Difference-in-Difference Regressions (2000-2008)

	Specification:		*Implied Treatment Effects:*	
Ages in Treatment Group:	(1)	(2)	%Δ	ATT
All Aged 50+	0.304*** (0.059)		36%	$1,068
Ages 50-59		0.187** (0.076)	21%	$740
Ages 60-69		0.404*** (0.062)	50%	$1,420
Ages 70-79		0.572*** (0.073)	77%	$1,592
Ages 80-89		0.540*** (0.115)	72%	$1,111
Number of Treated Individuals	406,496	406,496		
Number of Total Observations	7,228,503	7,228,503		

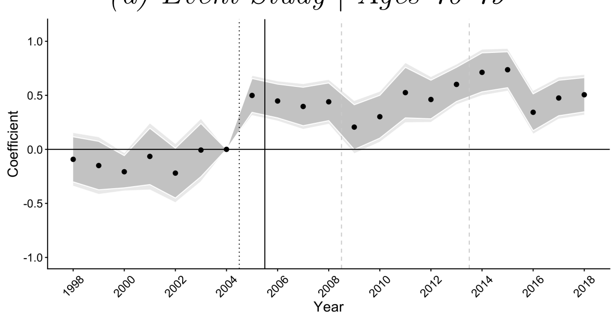
(b) Event Study | Ages 50-59



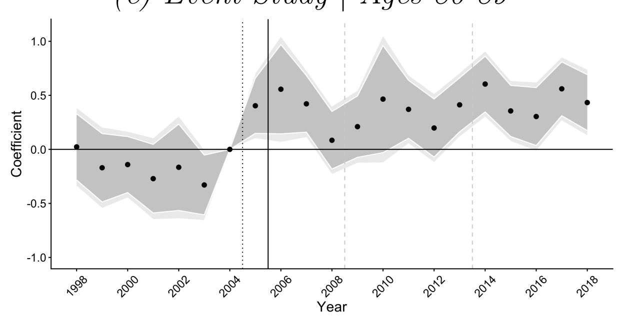
(c) Event Study | Ages 60-69



(d) Event Study | Ages 70-79



(e) Event Study | Ages 80-89



Note: This table presents estimates from our baseline empirical strategy which compares older age groups to a control group of individuals aged 30-49. Panel (a) estimates Equations 3 and 5 using Poisson Pseudo Maximum Likelihood. The outcome variable is net realized capital gains from any asset class, as reported yearly on each individual's tax return. Observations are at the individual-year level. Individuals aged 30 and above are included in these regressions if they ever owned stock. Mutually exclusive treatment indicator(s) equal 1 if the individual is in the indicated age group and equal 0 if the individual is in the 30-49 age group. Standard errors clustered at the individual level are presented in parentheses below. All regressions control for an indicator for private business ownership and a bin of highest-ever household net worth. Corresponding event study estimates are depicted in Panels (b)-(c). Individuals in these regressions are classified as private business owners if, since the age of 25, anyone in their household owned unlisted stock between 1993 and 2018. Individuals are assigned their household's highest net wealth percentile over the same period of time.

Table 3: Realization Responses by Age Group & Type of Capital Gain (2000†-2008)

ATT by Type of Capital Gains:	*Age Range of Treatment Group:* 50-59	60-69	70-79	80-89	All 50+
All	740**	1,420***	1,592***	1,255***	1,068***
Real Estate	-48	-77	-60	-122	-64*
Private Business	-76	58	-227	306*	-46
Listed Stock, Mutual Funds, & Other	537**	753***	1,020***	1,038***	695***

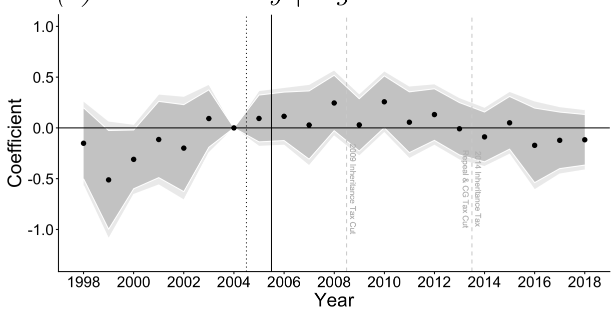
Note: This table presents the implied treatment effects (ATTs) from re-running Equations 3 and 5 using different outcome variables corresponding to capital gains from particular asset classes. The first row ("All") reproduces the same output as in Table 2, using the outcomes variable of net realized capital gains from any asset class, as reported yearly on each individual's tax return. Appendix Table A3 presents detailed regression tables underlying the other rows (other outcome variables). All regressions are estimated with Poisson Pseudo Maximum Likelihood at the individual-year level. Individuals aged 30 and above are included in these regressions if they ever owned stock. Mutually exclusive treatment indicator(s) equal 1 if the individual is in the indicated age group and equal 0 if the individual is in the 30-49 age group. All regressions control for an indicator for private business ownership and a bin of highest-ever household net worth.
†: Only our preferred outcome variable of "All" capital gains has full year coverage from 2000-2008. Capital gains for "Real Estate" are not observed in 2005 due to a data issue. Capital gains in "Private Business" and "Listed Stock, Mutual Funds + Other" are only observed from 2003 (when our stock transactions data begins). All analyses in this table end in 2008, before the 2009 decrease in inheritance tax rates.

Table 4: Realization Responses from Alternative (UCG-based) Empirical Design

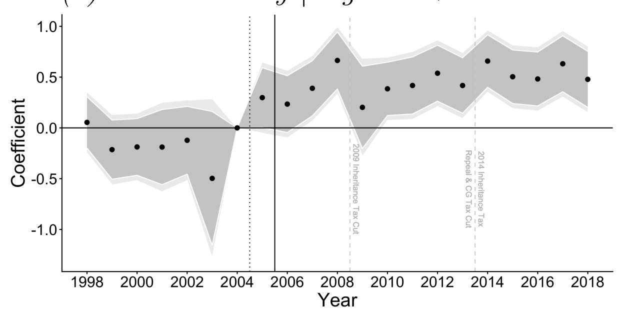
(a): Difference-in-Difference Regressions (2000-2008)

	Specification:		
	All	<50	50+
Indicator for $\gamma > 0$	0.430***	0.249***	0.532***
	(0.071)	(0.095)	(0.097)
Number of Treated Individuals	396,843	159,356	237,487
Number of Total Observations	4,526,325	1,827,559	2,698,766
Implied Treatment Effects			
Percent Change	54%	28%	70%
Average Increase (ATT)	\$1,791	\$1,131	\$2,110

(b): Event Study | Aged <50 in 2005

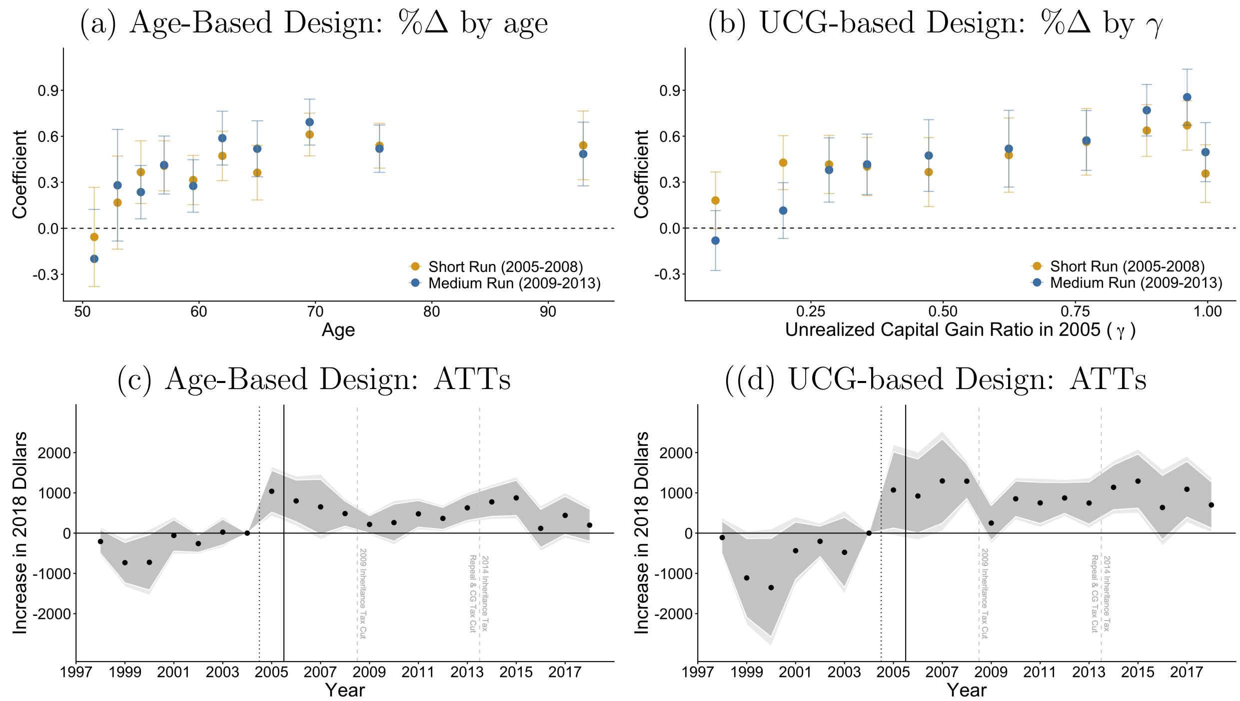


(c): Event Study | Aged 50+ in 2005



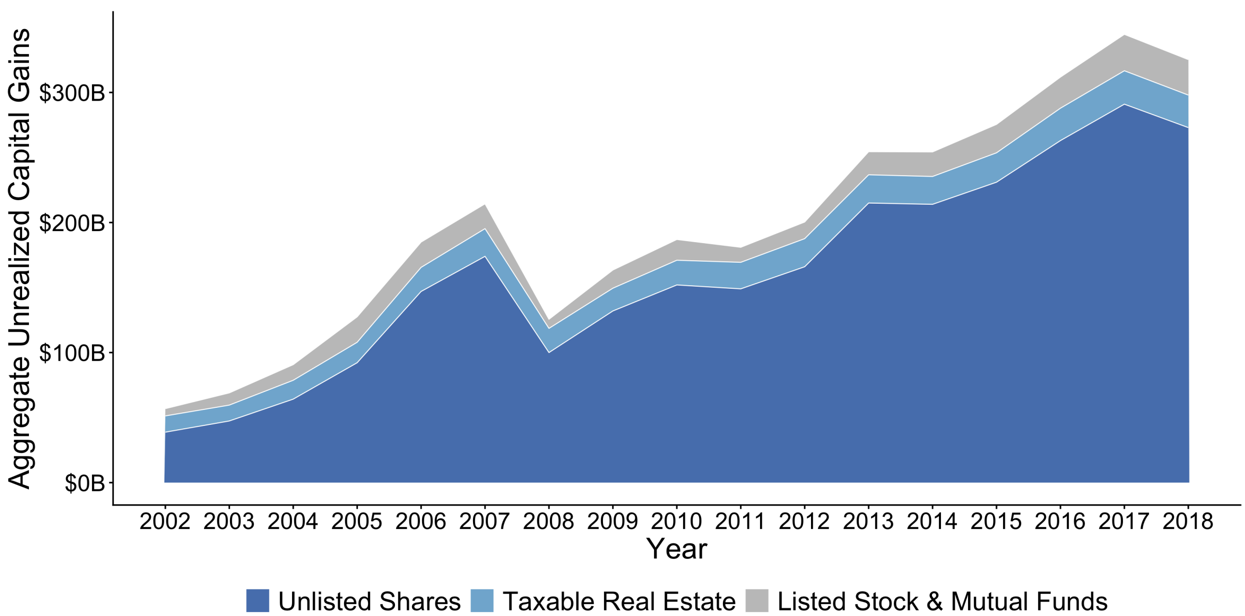
Note: This table displays estimates from our alternative empirical strategy which categorizes individuals as treated if they have positive appreciation in their stock porfolio at the beginning of 2005. Panel (a) presents binary difference-in-difference estimates of Equation 3, estimated using Poisson Pseudo Maximum Likelihood. The first column ("All") displays estimates for all individuals aged 30 and above who owned stock at the beginning of 2005. To increase comparability with the age-based design, the next two columns re-run the regression on the sample of stockowners in 2005 who were aged 30-49 ("<50") and aged 50 or above ("50+"). The outcome variable is net realized capital gains from any asset class, as reported yearly on each individual's tax return. All regressions include controls for age group, private business ownership, and age-specific rank in the household wealth distribution. Standard errors clustered at the individual level are presented in parentheses below. To interpret our coefficients in terms of percent changes in the outcome variable, we need to exponentiate the coefficient and subtract by one. We form an estimate of the ATT from regression parameters as described in Equation 4. Panels (b) and (c) display event study estimates broken out by age group (corresponding to the difference-in-difference estimates of columns 2 and 3 of Panel (a)). Appendix Figure A3 displays event study estimates broken out by treatment intensity (i.e., different levels of positive γ).

Figure 4: Heterogeneity & Aggregation, for Two Empirical Strategies



Note: This figure shows realization responses over time for our two empirical strategies. Our baseline age-based strategy compares older individuals (aged 50+) to a control group of individuals aged 30-49. The alternative UCG-based strategy compares individuals with more capital gain in their stock portfolio in 2005 to a control group of individuals with zero or negative capital gain in stock in 2005. All regressions control for age-adjusted wealth percentile and whether an individual owns a private business. The UCG-based regressions additionally control for age. Panels (a) and (b) show realization responses broken out by time period and the intensity of treatment (decile of age and decile of positive capital gains ratio, respectively). Treatment effects and 95 percent confidence intervals for each decile of treatment are estimated in a diff-in-diff regression following Equation 5. Panels (c) and (d) present Average Treatment Effects (ATTs in 2018 USD) for the binarized treated group estimated from Equation 3. The event study estimates are binary in the sense that they compare all "treated" units within the empirical strategy to the control group. Light gray shading indicates 95 percentile confidence intervals and dark gray shading indicates 90th percentile confidence intervals. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. The dashed gray line before 2009 indicates a fall (by half) of the statutory inheritance tax rate. The dashed gray line before 2014 indicates the full removal of the inheritance tax, and the beginning of a period of changing capital gains tax rates. See Appendix F for more detail on these later policy changes.

Figure 5: Unrealized Capital Gains Over Time, 2002-2018



Note: This figure shows aggregate unrealized capital gains in taxable asset categories at the end of each year from 2002 (the beginning of our stock transactions data) until 2018. Aggregate unrealized capital gains have mostly grown over time, with the exception of a sharp drop during the Great Recession. Trends are entirely driven by unrealized capital gains in unlisted stock. The "Listed Stock & Mutual Funds" category in this graph includes the upper bound assumption about unrealized capital gains in mutual funds.

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Appendix

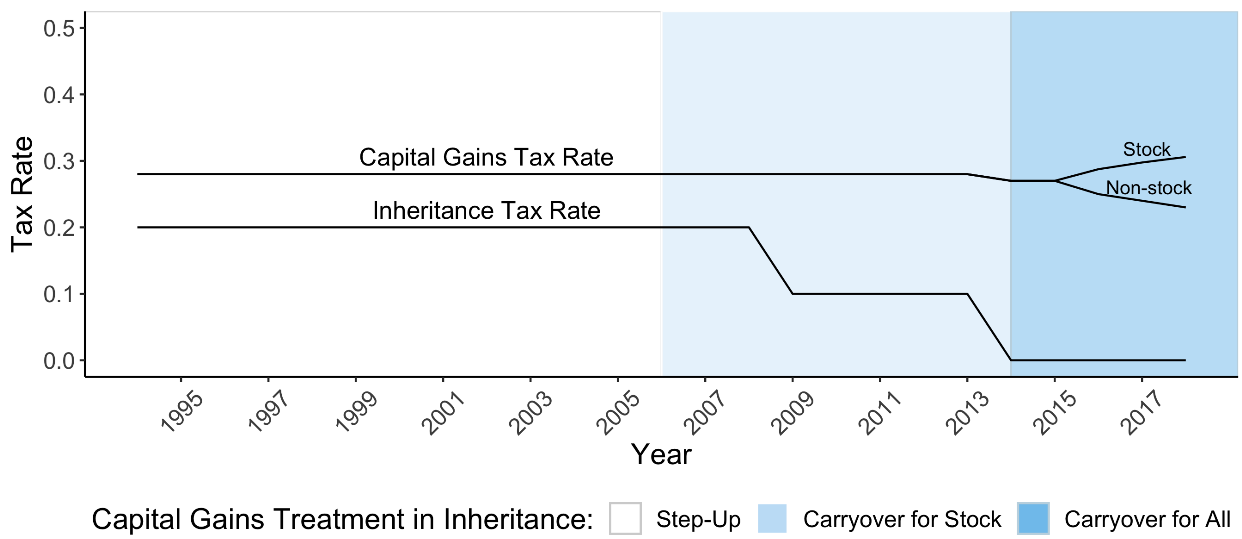
A Additional Appendix Tables & Figures

Table A1: Data Sources and Their Role in the Measurement Exercise

Data source	Coverage	Unit of observation	Key variables	Role in measurement
Panel A. Core administrative records				
Annual tax records: income and wealth variables	1992–2018	Household-year	Asset holdings (mutual funds, property, listed and unlisted stocks), liabilities, taxable income	Measures household wealth, wealth ranks, portfolio shares, and mutual-fund wealth.
Stock transactions	2003–2018 + last pre-2003 transaction	Individual-security-transaction	Asset ID, buyer/seller, quantity, price, transfer type	Reconstructs stock holdings, basis, RCG, UCG, and ICG for listed and unlisted stock.
Real estate transactions	1993–2018 + last pre-1993 transaction	Individual-property-transaction	Property ID, buyer/seller, price, transfer type	Reconstructs real-estate holdings, basis, RCG, UCG, and ICG for real estate.
Panel B. Valuation and validation inputs				
Unlisted firm accounts (financial and tax-accounting records)	1993–2018	Firm-year	Sales, income, book values, industry, ownership links	Constructs market value estimates for unlisted firms.
Shareholder Registry	2004–2018	Individual-company-year	End-of-year shareholdings	Validates the transaction-based stock ledger.

Note: This table summarizes the main data sources used in the measurement exercise. Listed stock values are obtained from Thomson-Reuters Datastream. Real estate values use the property-level market-value estimates from Eika et al. (2020). Mutual fund wealth is observed in the wealth data, but mutual fund transactions and basis are not observed; we therefore bound mutual-fund UCG as described in Section 1.3. Stock and real estate transaction records include sales, inheritance transfers, and other transfer types. Data limitations are discussed in Section 1.3.

Figure A1: Capital Gains and Inheritance Policy in Norway, 1994–2018



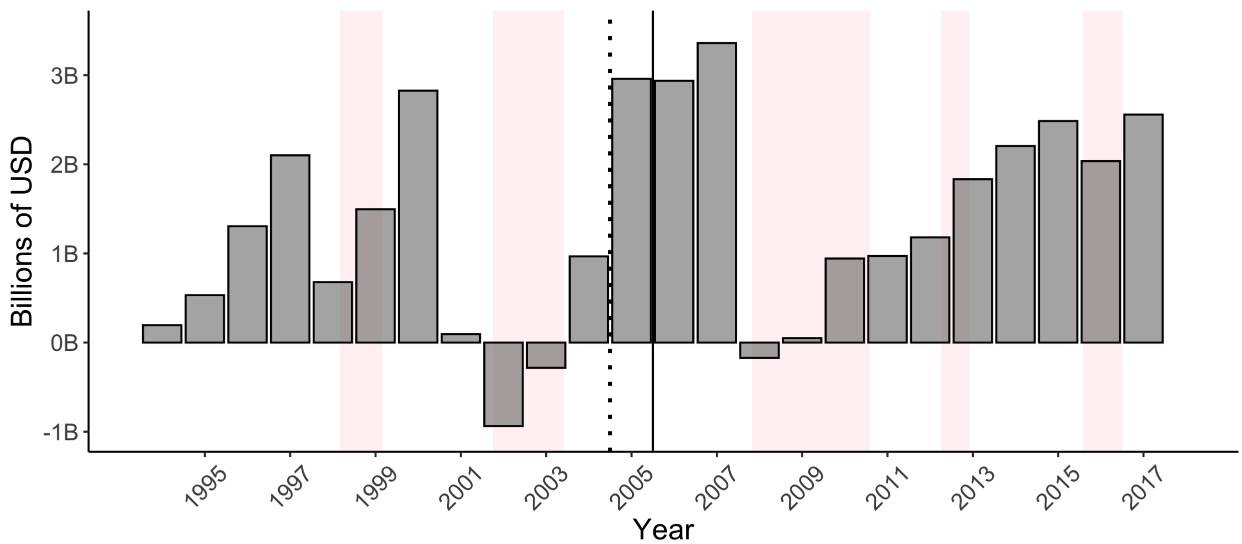
Note: This figure summarizes the changes to capital gains and inheritance taxation that took place during our study period. The statutory capital gains tax rate on all taxable assets was steady at 28% until 2014. From 2016, separate capital gains tax rates were introduced for stock and non-stock assets. Throughout our period, capital gains taxation in Norway has had a flat rate structure, although the rate was made to depend on the asset type (stock or non-stock) in later years. In contrast, inheritance tax rates (prior to their abolition) were progressive in the cumulative amount of the inheritance received, and depended on the relationship between giver and recipient. This graph depicts the top statutory inheritance tax rate for closely related family members, which fell from 20% to 10% in 2009 and to 0% with the abolition of the inheritance tax in 2014. See Appendix Table A2 for more information on inheritance tax rates and brackets. The shading of the graph depicts changes to the treatment of inherited capital gains. Prior to 2006, all gifted and inherited assets benefited from stepped-up basis. From 2006, gifted and inherited stock with capital gains were switched to a system of carryover basis. From 2014, all gifted and inherited assets carried their basis over to the recipient, unless the asset would not have been subject to capital gains taxation in the hands of the original owner.

Table A2: Inheritance Tax Schedule, 1985-Present

Years	Amount Inherited (NOK)	Tax Rate for:		
		Spouses	Close Relatives	Other Recipients
1985-1998	0 - 100,000	0	0	0
	100,000 - 400,000	0	0.08	0.10
	400,000 +	0	0.20	0.30
1999-2002	0 - 200,000	0	0	0
	200,000 - 500,000	0	0.08	0.10
	500,000 +	0	0.20	0.30
2003-2008	0 - 250,000	0	0	0
	250,000 - 550,000	0	0.08	0.10
	550,000 +	0	0.20	0.30
2009-2013	0 - 470,000	0	0	0
	470,000 - 800,000	0	0.06	0.08
	800,000 +	0	0.10	0.15
2014-	Any	0	0	0
		0	0	0
		0	0	0

Note: This table depicts the Norwegian inheritance tax scheme from 1985 until the inheritance tax was abolished in 2014. Inheritance tax was levied without distinction between *inter vivos* gifts and bequests at the time of death. The rate structure depended on the cumulative amount of the gift/inheritance received from the individual giver, and the relationship between the giver and the recipient. The definition of close relatives includes parents and children of the giver. Transfers between spouses were not taxed. Transfers between individuals who are not close relatives were subject to a higher top rate in every year. In 2014, the gift and inheritance tax was abolished, so all rates for subsequent years are set to 0%.

Figure A2: Aggregate Net Capital Gains Realizations by Year, 1994-2018



Note: This figure depicts the total amount of net capital gains (i.e., capital gains minus capital loss) realized in Norway from 1994 to 2018, in terms of 2018 USD. The light pink shading denotes recessionary periods, as classified by the OECD. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. Net capital gains realizations tripled between 2004 and 2005: from \$0.97 billion in 2004 (6.24 billion nominal NOK) to \$2.96 billion in 2005 (19.4 billion nominal NOK).

Table A3: Realization Responses by Age Group & Specified Type of Capital Gain

Panel (a): Real Estate Capital Gains (2000-2008)

	(1)				(2)
Age Range of Treatment Group:	50-59	60-69	70-79	80-89	All 50+
Group × POST	-0.087	-0.106	-0.093	-0.300	-0.105*
	(0.068)	(0.077)	(0.108)	(0.183)	(0.057)
Number of Observations	6,372,678	6,372,678	6,372,678	6,372,678	6,396,882
Implied ATT	$-48	$-77	$-60	$-122	$-64

Panel (b): Private Business Capital Gains (2003-2008)

	(1)				(2)
Age Range of Treatment Group:	50-59	60-69	70-79	80-89	All 50+
Group × POST	-0.043	0.036	-0.211	0.497*	-0.031
	(0.172)	(0.175)	(0.386)	(0.261)	(0.140)
Number of Observations	4,975,814	4,975,814	4,975,814	4,975,814	5,002,319
Implied ATT	$-76	$58	$-227	$306	$-46

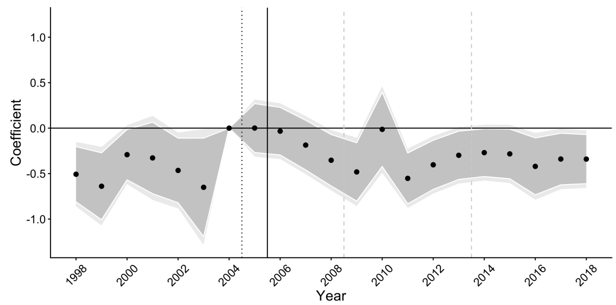
Panel (c): Listed Stock, Mutual Funds & Other Capital Gains (2003-2008)

	(1)				(2)
Age Range of Treatment Group:	50-59	60-69	70-79	80-89	All 50+
Group × POST	0.234**	0.360***	0.564***	0.702***	0.331***
	(0.103)	(0.094)	(0.119)	(0.173)	(0.083)
Number of Observations	4,117,588	4,117,588	4,117,588	4,117,588	4,135,958
Implied ATT	$537	$753	$1,020	$1,038	$695

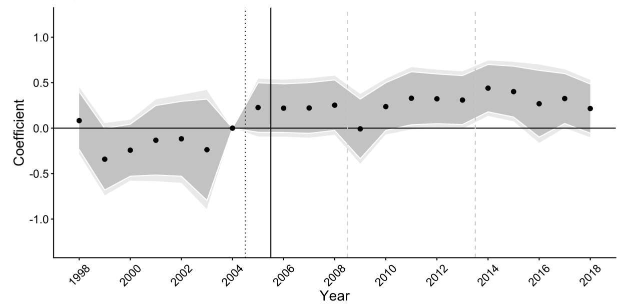
Note: This table presents difference-in-difference estimates from our age-based empirical strategy (as in Tables 2 and 3), but with the outcome variable in each panel corresponding to particular types of capital gain. All regressions include covariates for whether the individual ever owns a private business and for their highest age-net worth bin. Observations are at the individual-year level. Individuals are included in these regressions if they owned stock at the end of 2004. Mutually exclusive treatment indicator(s) equal 1 if the individual is in the age group below the numerical column header, and equal 0 if the individual is in the 30-49 age group. Other age groups are not included. The indicator "POST" equals 1 for years after 2004, and 0 before then. Standard errors clustered at the individual level are presented in parentheses below. To interpret the coefficients in terms of percent changes in the outcome variable, exponentiate the coefficient and subtract by one. We form an estimate of the ATT from regression parameters as described in Equation 4. ATT values are in 2018 USD. Capital gains for "Real Estate" are not observed in 2005 due to a data issue. Capital gains in "Private Business" and "Listed Stock, Mutual Funds + Other" are only observed from 2003 (when our stock transactions data begins). All analyses in this table end in 2008, before the 2009 decrease in inheritance tax rates. Individuals in these regressions are classified as private business owners if, since the age of 25, anyone in their household owned unlisted stock between 1993 and 2018. Individuals are assigned their household's highest net wealth percentile over the same period of time.

Figure A3: Event Studies for the UCG-based Design, by Bin of Treatment Intensity

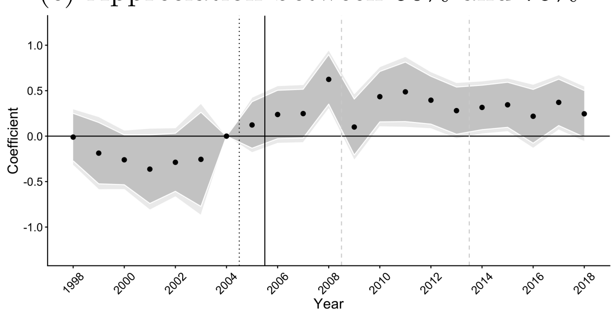
(a) Appreciation between 0 and 25%



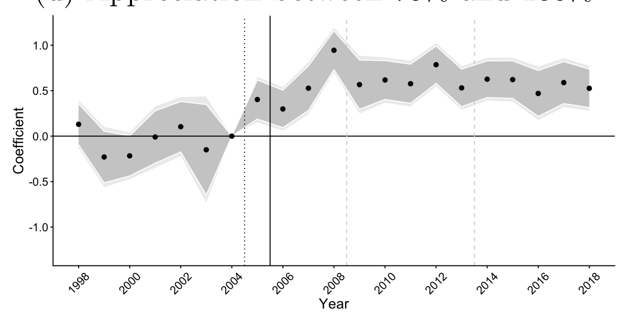
(b) Appreciation between 25% and 50%



(c) Appreciation between 50% and 75%

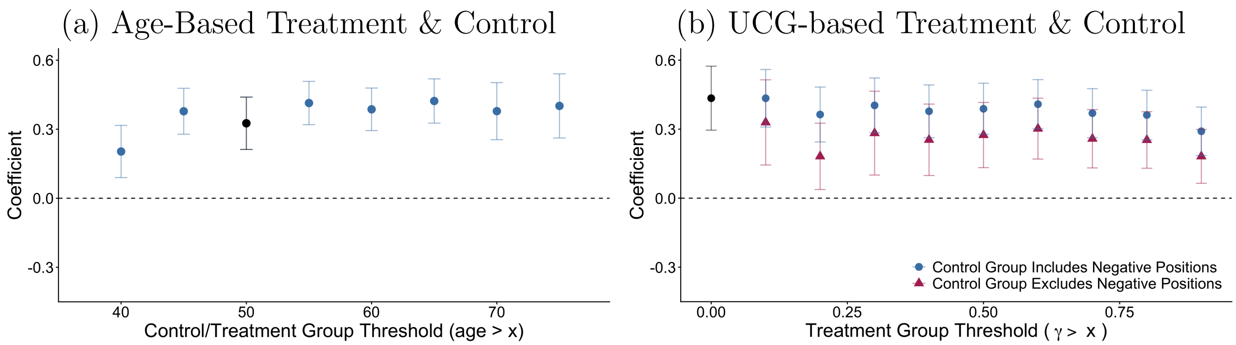


(d) Appreciation between 75% and 100%



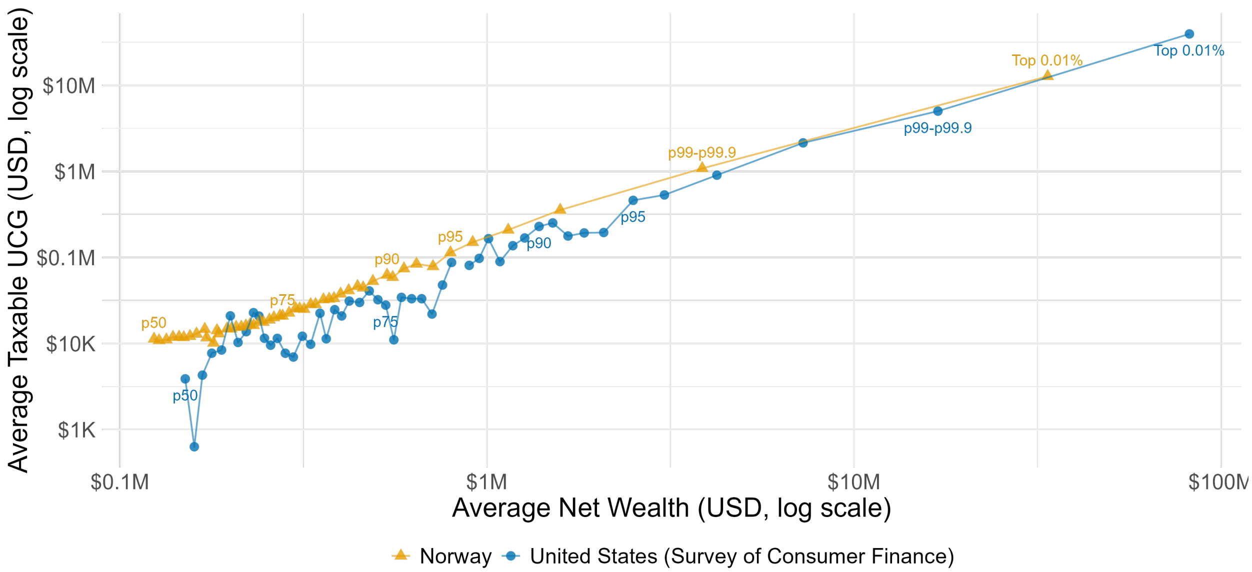
Note: This figure shows heterogeneity in the average responses to the removal of stepped-up basis for inherited stock. Panel (a) depicts event study coefficients for individuals whose stock holdings, at the end of 2004, have a ratio of unrealized capital gains to value (γ) between zero and 0.25. Panel (b) shows the corresponding figure for individuals with 2004 stock holdings with γ between 0.25 and 0.5. Panel (c) is the corresponding figure for those with γ between 0.5 and 0.75. Panel (d) is the corresponding figure for those with γ between 0.75 and 1. In all panels, the control group is individuals whose stock portfolios in 2005 had zero capital gains or net capital losses and the outcomes variable is net realized capital gains from any asset class. All regressions include controls for age group, private business ownership, and age-specific rank in the household wealth distribution. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. The dashed gray line before 2009 indicates a fall (by half) of the statutory inheritance tax rate. The dashed gray line before 2014 indicates the full removal of the inheritance tax, and the beginning of a period of increasing capital gains tax rates.

Figure A4: Robustness to Alternative Control Group Definitions



Note: This figure shows how our binary estimates of realization responses from Tables 2 and 4 change when we alter the threshold at which we define our "treatment" and "control" groups. Each point is a difference-in-difference estimate of the binarized specification of Equation 3 with the treatment variable defined as an indicator variable for age or γ above the given threshold. The threshold in Panel (a) is defined in terms of age (baseline threshold: age $>$ 50 years) and the threshold in Panel (b) is defined in terms of the individual's unrealized capital gain ratio (γ) in 2005 (baseline threshold: $\gamma > 0$). The red triangles in Panel (b) additionally show how results change when individuals with negative positions are excluded from the control group, so that the control group consists only of individuals with non-negative gains up to the threshold. As discussed in Section 5, both of our empirical designs are "continuous" or "dose-response" difference-in-differences designs, and neither have an obvious cutoff threshold at which to cleanly separate the treatment and control group. Even when true treatment effects increase monotonically with the running variable, incremental increases in the choice of threshold may increase or decrease estimated effects. Increasing the treatment threshold in the treated range misattributes some of the true causal effect to the control group mean but also re-defines the treatment group to only include more intensely treated members. Which mechanism dominates at intermediate values will depend on the slope of the heterogeneous treatment effects by dosage.

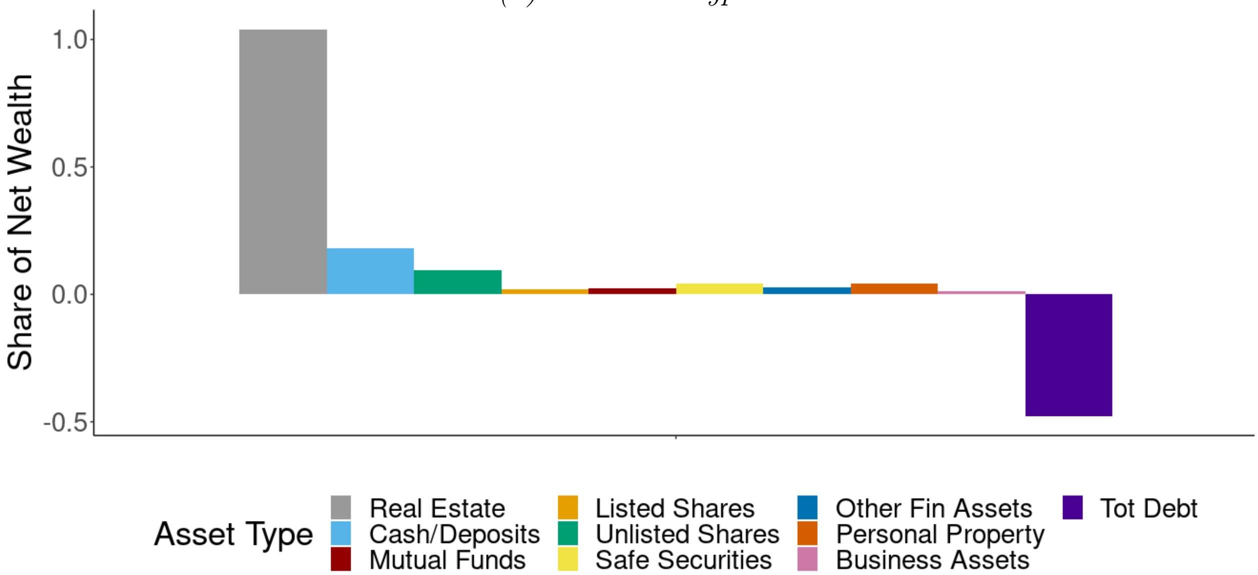
Figure A5: UCG vs. Net Wealth: Comparing Norway and the United States, 2004



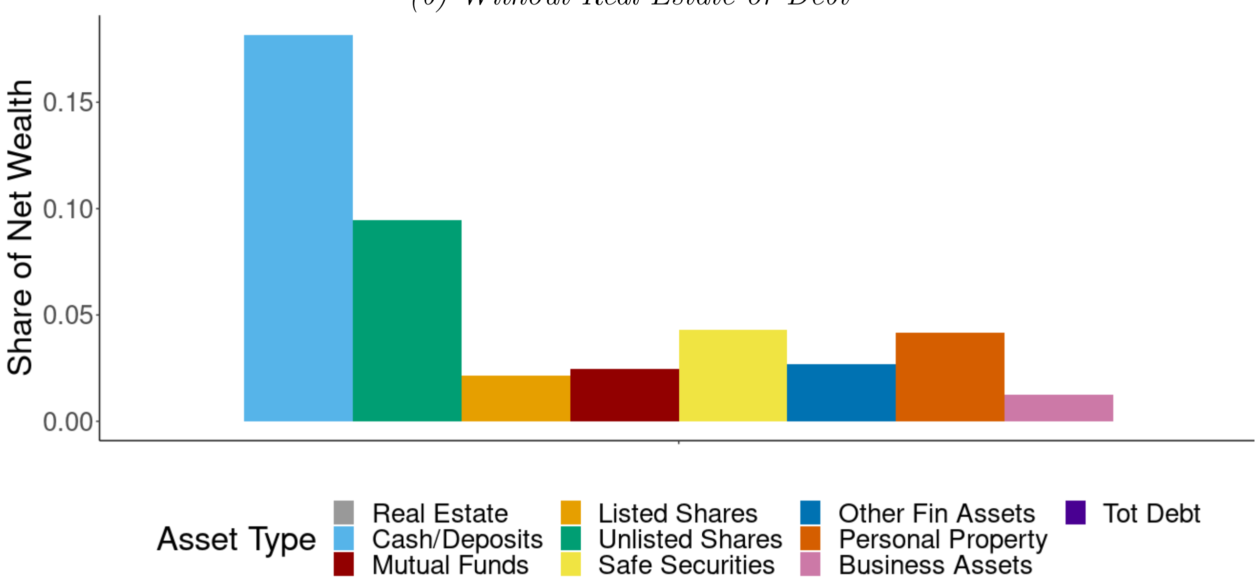
Note: This figure compares our measure of unrealized capital gains in taxable asset categories (i.e., excluding non-primary real estate) in Norway at the end of 2004 to unrealized capital gains in categories other than primary residences reported in the 2004 U.S. Survey of Consumer Finance. Each point in the figure represents a mean for individuals grouped (within their respective data source) by total net worth. We group individuals at integer percentiles of net worth below the 99th percentile, but then break the 99th percentile into a group below the 99.9th percentile (labeled "p99-p99.9") and a group of the Top 0.01%. Middle-class Norwegians have comparable but slightly higher levels of unrealized capital gains than middle-class U.S. residents with the same amount of total wealth. Average unrealized capital gains converge for U.S. and Norwegian millionaires. At the very top, the top of the U.S. wealth distribution extends past the range of the Norwegian distribution, so it is not possible to directly compare magnitudes.

Figure A6: Average Portfolio Shares in 2004

(a) All Asset Types



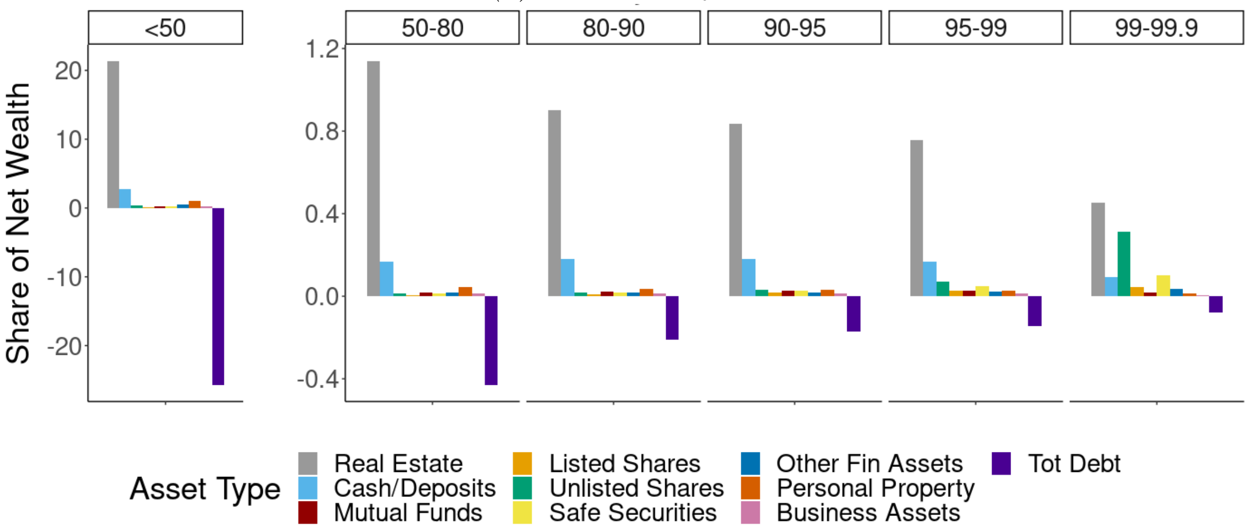
(b) Without Real Estate or Debt



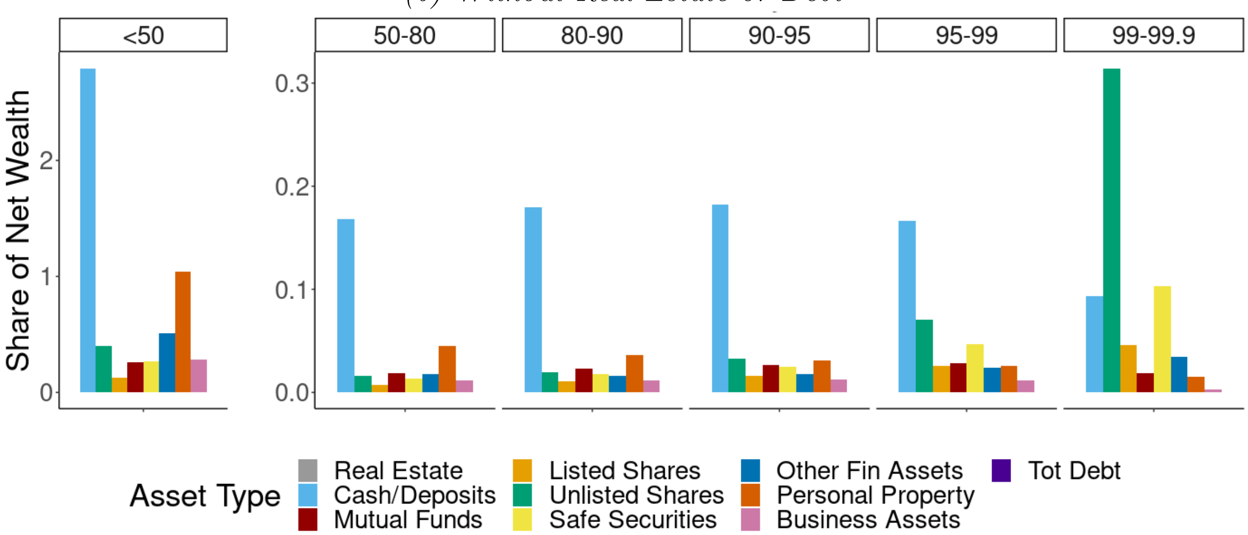
Note: This figure depicts portfolio shares for the average Norwegian household at the end of 2004. Panel (a) presents all asset types. Panel (b) is a zoomed in version for those asset types other than real estate and debt.

Figure A7: Average Portfolio Shares in 2004, by Net Worth Percentile

(a) All Asset Types



(b) Without Real Estate or Debt



Note: This figure depicts portfolio shares for Norwegian household broken out by percentile of net worth at the end of 2004. Panel (a) presents all asset types. Panel (b) is a zoomed in version for those asset types other than real estate and debt. Note that averages below the 50th percentile include people with negative net worth (lots of debt), implying portfolio shares greater than 1. Unlisted stock is a disproportionately important part of very rich people's portfolios – people between the 99th and 99.99th percentile of net wealth in 2004 had an average holding of $1.3 million (31% of net wealth).

B Model Simulation & Extension

This section provides implementation details of how we numerically solve & simulate the model from Section 4 and details how we extend the model to allow individuals to make early transfers during the 2005 anticipation period.

B.1 Model Simulation Details

B.1.1 Parameterizations

We parameterize the utility function

$$u(c_t) \equiv \tfrac{1}{1-\rho} c_t^{1-\rho} \qquad \nu(B_{\mathcal{T}}) \equiv \tfrac{\beta}{1-\beta}\tfrac{1}{1-\rho}\big(A \cdot B_{\mathcal{T}}\big)^{1-\rho}$$

where β is the discount factor, ρ is the relative risk aversion coefficient and $A \equiv \frac{r-i}{(1+i)}$ is the individual's utility weight on bequests. Note that this formulation of the utility over bequests is equivalent to assuming that the proceeds of the inheritance are used to purchase a perpetual annuity which provides the heir with a stream of consumption $A \cdot B_{\mathcal{T}}$.

B.1.2 Recursive Formulation

We can write the Bellman for state variables $X_t = [P_t, U_t, n_{t-1}, W_t]'$ as

$$V_t(X_t;\theta) = \mathbb{P}(\mathcal{T} > t | \mathcal{T} \geq t) \max_{c_t, n_t \geq 0} \big\{ u\big(c_t\big) + \beta E_t[V_{t+1}(X_{t+1};\theta)]\big\} + \mathbb{P}(\mathcal{T} = t | \mathcal{T} \geq t)\nu\big(B_t\big) \quad s.t.$$

$$W_{t+1} = \big(1+r\big)\big(W_t - n_t P_t - c_t - \tau_c R_t\big) + n_t P_{t+1}$$

$$B_t = W_t(1-\tau_I) - \theta \tau_c U_t^{+}$$

B.1.3 Normalized Formulation

Our choice of parameterization (with constant relative risk-aversion) implies that optimal consumption and portfolio choices will be independent of wealth. Following Dammon et al. (2001)'s methodology, we solve a transformation of the above model that normalizes all of the variables by beginning-of-period wealth. With beginning-of-period wealth W_t as the numeraire, we can write a normalized value function $v_t(x_t;\theta) \equiv V_t(X_t;\theta)/W_t^{1-\rho}$

$$v_t(x_t;\theta) = \mathbb{P}(\mathcal{T} > t|\mathcal{T} \geq t) \max_{c_t, n_t \geq 0} \left\{ u\big(\tfrac{c_t}{W_t}\big) + \beta E_t\Big[v_{t+1}(x_{t+1};\theta)\big(\tfrac{W_{t+1}}{W_t}\big)^{1-\rho}\Big] \right\} + \mathbb{P}(\mathcal{T} = t|\mathcal{T} \geq t)\, \tilde{\nu}(x_t;\theta)$$

with the normalized death payoff:

$$\tilde{\nu}(x_t;\theta) = \frac{\beta}{1-\beta}\frac{1}{1-\rho}\big(A(1-\tau_I - \theta\tau_c s_t \gamma_t^+)\big)^{1-\rho}$$

This formulation only requires two (normalized) state variables: $x_t = [s_t, \gamma_t]$, where $s_t \equiv \frac{n_{t-1}P_t}{W_t}$ is the beginning-of-period equity share of wealth and $\gamma_t \equiv U_t/s_t W_t$ is the beginning-of-period unrealized capital gains share of equity. $\gamma_t^+ = \max\{0, \gamma_t\}$.

B.1.4 Simulation

The model cannot be solved analytically, so we use numerical techniques to solve for the policy and value functions. We employ our (numerical) policy and value functions in a simulation which assumes that every individual starts life at age 20 ($t = 0$) with $W_0 = 1$ and no unrealized capital gains ($U_0 = 0$). Individuals live up to age 100 ($T = 80$) and face a yearly probability of death based on official Norwegian mortality numbers (Statistics Norway, 1966–2024). Table A4 displays our chosen values that further parameterize the model for simulation purposes. At advanced ages the value function is nearly flat in the portfolio choice (within 3×10^{-4} of the optimum over a ± 0.05 range of s), so the solved policy is identified only up to numerical noise within this indifference set; for figures we select from that set via a centered three-age moving average, which changes lifetime value by less than the indifference tolerance and mean realizations by under 1%,

Table A4: Parameters for Model Simulation

Parameter	Symbol	Value	Source	Timeframe
Financial markets				
After-tax risk-free rate	$r \equiv (1-\tau_d)r_f$			
Pretax risk-free rate	r_f	0.0376	Fagereng et al. (2017) (using Klovland (2004))	1900-2005
Mean capital gain return	$\bar{g}$	0.0697	Fagereng et al. (2017) (using Dimson et al. (2008), Klovland (2004))	1900-2005
Std. dev. of capital gains	σ	0.244		
Inflation rate	i	0.023	Grytten (2004)	1995-2003
Tax rates				
Tax rate on interest	τ_d	0.28	Norwegian Law (flat rate)	
Tax rate on realized capital gains	τ_c	0.28	Norwegian Law (flat rate)	1992-2013
Inheritance tax rate	τ_I	0.20	Norwegian Law (top rate)	1985-2013
Preferences				
Discount factor	β	0.96		
Coefficient of relative risk aversion	ρ	3		
Lifecycle				
Starting age	$t=0$	20		
Maximum age	T	100		
Wealth at starting age	W_0	1		
Unrealized Gains at start	γ_0	0		
Mortality data source			Statistics Norway (1966–2024)	2004

Note: This table displays our chosen values that parameterize the model for simulation purposes. The "Financial markets" parameters are chosen to follow Norwegian portfolios in the decades prior to our policy change and assume that Norwegian equity portfolios are weighted 80% to Norwegian equity, 20% to global equity (following Fagereng et al. (2017)). "Tax rates" are chosen to be consistent with actual Norwegian law around the time of our policy change. Table A5 provides additional detail on how we transform the equity-related parameters from the literature into the (nominal) values listed here. Table A2 provides additional details on inheritance tax rates.

Table A5: Calculation of Parameters for Model Simulation

Parameter	Real Value	Nominal Conversion	Source
Mean Equity Returns			
Norway	0.0428		Dimson et al. (2008)
World	0.0575		Dimson et al. (2008)
80/20 Weighted Portfolio	0.0457	0.0697	Fagereng et al. (2017)
Std. dev. of Equity Returns			
Norway	0.2696		Dimson et al. (2008)
World	0.1723		Dimson et al. (2008)
corr(Norway, World)	0.61		Ødegaard (2007)
80/20 Weighted Portfolio	0.238	0.244	Fagereng et al. (2017)

Note: This table explains how we transform values taken from the literature into the values used in our simulation purposes. We employ Fagereng et al. (2017)'s estimate (combining administrative data on directly-held stock with aggregate data on mutual fund holdings) that Norwegian equity portfolios including mutual funds are weighted 80% Norway, 20% the rest of the world. Real values are transformed into nominal values using an inflation rate of 2.3% which Grytten (2004) finds to be the average inflation rate in Norway for 1995-2003. Norges bank adopted an inflation target of 2.5% in 2001.

B.2 Model Extension for Anticipatory Transfers

Our baseline model does not include the possibility of *inter-vivos* gifts: all intergenerational transfers are deathtime bequests. This significantly simplifies the model and is consistent with the fact that, before and after the 2005-2006 change in Norway, most intergenerational transfers take place in the year or two prior to death. However, 2005 was a unique year for intergenerational transfers. Throughout 2005, Norwegians were aware that intergenerational transfers (both gifts and bequests) given this year would be subject to step-up, but that transfers in future years would be subject to carryover basis treatment. Accordingly, there was a clear incentive in 2005 to give early transfers, and there was a large spike in 2005 in *inter-vivos* giving. To allow for these anticipatory gifts in the model, we modify that model so that in a single "announcement year" (t_a), individuals have the opportunity to set aside some of their current equity to guarantee step-up treatment for that portion (stepped up to current value at the time of the announcement). After this commitment decision, they will have to set that precommitted equity aside and it cannot be used for their own consumption or rebalancing. To reflect the fact that early giving is typically disfavored, we assume the individual does not receive utility from the inter-vivos gift until the normal bequest time and does not value any appreciation that occurs in the asset between t_a and $\mathcal{T}$.

B.2.1 Timeline and Regimes

1. **Pre-announcement** $(t < t_a)$**:** Individuals behave according to step-up equilibrium, unaware of coming policy change

2. **Announcement period** $(t = t_a)$**:** Policy change is announced for future implementation. Individuals make a precommitment decision about how much equity to pass to heirs with step-up basis. After this commitment decision, they will have to set that precommitted equity aside and it cannot be used for their own consumption or rebalancing.

3. **Post-announcement** $(t > t_a)$**:** Individuals follow a dynamic program with three state variables, tracking both their current portfolio and their precommitted bequest amount

4. **At Death** $(t = \mathcal{T})$ At the time of death, the individuals' heir receives both the precommitted bequest (with basis stepped-up to the time of the precommitment) and any additional terminal wealth (with carryover basis).

B.2.2 Modeling the Anticipation Decision

At the announcement time t_a, individuals have a one-time choice to commit some of their current equity as a "precommitted bequest". The individual chooses $B_{\text{pre},t_a} \leq s_{t_a}$ to solve a one-time decision:

$$v_{t_a}^{\text{pre}}(X_{t_a}) = \max_{c_{t_a}, \quad n_{t_a} \geq 0, \quad B_{\text{pre},t_a} \leq s_{t_a}} \left\{ u(\frac{c_{t_a}}{W_{t_a}}) + \beta \cdot E[v_{t_a+1}\left[s_{t_a+1}, \gamma_{t_a+1}, B_{\text{pre},t_a}\right](\frac{W_{t_a+1}}{W_{t_a}})^{1-\rho}] \right\}$$

subject to a modified budget constraint:

$$W_{t_a+1} = W_{t_a} + (P_{t_a+1} - P_{t_a})n_{t_a} - c_{t_a} - \tau_c G_{t_a} - B_{pre,t_a} W_{t_a}$$

where $B_{pre,t_a} \leq s_{t_a}$ is the fraction of W_{t_a} that is precommitted.

In the post-announcement periods ($t > t_a$) there are 3 state variables:

- $s_t = \frac{n_{t-1}P_t}{W_t}$ the currently available equity share (i.e., excluding B_{pre})
- γ_{t_a+1} the current ratio of unrealized capital gains to value in the available equity
- $B_{\text{pre},t}$ the ratio of precommitted value to wealth in period t (new)

The decision variables are as before:

$$v_t^{\text{pre}}(X_t) = \max_{c_t,\quad n_t \geq 0} \left\{ u(\frac{c_t}{W_t}) + \beta \cdot E[v_{t+1}\left[s_{t+1}, \gamma_{t+1}, B_{\text{pre},t+1}\right](\frac{W_{t+1}}{W_t})^{1-\rho}] \right\}$$

The terminal value at time $\mathcal{T}$ makes sure that the precommitted equity is not subject to capital gains tax on appreciation prior to t_a:

$$v_{\mathcal{T}}^{\text{pre}} = \frac{\beta}{1-\beta} \times \frac{\left[A \times (1 + B_{pre,\mathcal{T}} - (\tau_I(1 + B_{pre,\mathcal{T}}) + \tau_c s_{\mathcal{T}} \gamma_{\mathcal{T}})^+)\right]^{1-\rho}}{1-\rho} \tag{9}$$

where $B_{\text{pre},\mathcal{T}}$ is the locked-in early bequest amount relative to $W_{\mathcal{T}}$. Note that $B_{\text{pre},\mathcal{T}}$ is subject to inheritance but not capital gains tax. Inheritance from terminal (non-precommitted) equity ($s_{\mathcal{T}}$) is subject to capital gains tax. This modification of the terminal value creates a tradeoff for individuals at the announcement time: precommiting more equity raises the terminal value but less precommitment allows individuals to maintain liquidity flexibility for consumption/rebalancing and to benefit from appreciation that occurs between t_a and T.

C MVPF Derivation

C.1 MVPF Derivation for a Marginal Policy Change $d\theta$

We derive our equation for the MVPF under a marginal policy change (Equation 7) from the decision of an individual entering the period before death with positive unrealized capital gains ($U_{\mathcal{T}-1} > 0$). Assuming that this individual makes his final choice of stock holdings $n_{\mathcal{T}-1} \in [0, n_{\mathcal{T}-2}]$ and that his heir realizes inherited gains immediately upon receipt (before any additional price appreciation), we can simplify the terminal period budget constraint (from Equation 1) to $W_{\mathcal{T}} = W_{\mathcal{T}-1} - c_{\mathcal{T}-1} - \tau_c U_{\mathcal{T}-1}(1 - \frac{n_{\mathcal{T}-1}}{n_{\mathcal{T}-2}})$.

Then for any $\theta \in [0, 1]$ (i.e., step-up, carryover, or anything in between), we can write affected government revenue as: $G(\theta) \;\; = \tau_I W_{\mathcal{T}}(\theta) + \theta \tau_c U_{\mathcal{T}}(\theta) + \tau_c(1 - \frac{n_{\mathcal{T}-1}(\theta)}{n_{\mathcal{T}-2}})U_{\mathcal{T}-1}$

Plugging in $U_{\mathcal{T}-1}\frac{n_{\mathcal{T}-1}(\theta)}{n_{\mathcal{T}-2}} = U_{\mathcal{T}}(\theta)$, we differentiate $G(\theta)$ with respect to θ to recover the change in net government revenue for any marginal policy change $d\theta$:

$$dG(\theta) = \tau_I dW + \tau_c U_{\mathcal{T}}(\theta)d\theta + \tau_c(\theta - 1)\frac{U_{\mathcal{T}-1}}{n_{\mathcal{T}-2}}dn$$

Define $R_{\mathcal{T}-1} \equiv U_{\mathcal{T}-1} - U_{\mathcal{T}}$. From Equation 2, this implies $\frac{dR_{\mathcal{T}-1}}{d\theta} = -\frac{U_{\mathcal{T}-1}}{n_{\mathcal{T}-2}}\frac{dn_{\mathcal{T}-1}}{d\theta}$. Using this to substitute in for dn and the per-period budget constraint to substitute in for dW, we get:

$$dG(\theta) = -\tau_I dc + \tau_c U_{\mathcal{T}}(\theta)d\theta + \tau_c\big(1 - \theta - \tau_I\big)\frac{dR_{\mathcal{T}-1}}{d\theta}$$

This is the denominator of the MVPF for a marginal policy change. We will additionally assume $dc \approx 0$ (i.e., the older generation's marginal propensity to consume out of a terminal period wealth shock is zero).[59] The numerator of the MVPF is the dynasty's willingness-to-pay to avoid carryover policy, which is equal to their tax savings under step-up $WTP(\theta) = \tau_c U_{\mathcal{T}}(\theta)d\theta$. Divide through to get $MVPF_{d\theta} = \frac{WTP(\theta)}{dG(\theta)} = \frac{1}{1+FE_{d\theta}}$ for Equation 7.

C.1.1 Aggregation to the Discrete Reform

The removal of step-up is a discrete move from $\theta = 0$ to $\theta = 1$, so we aggregate the marginal effects by integrating along the policy path (Kleven, 2021). The MVPF of the reform is the ratio of the integrated willingness-to-pay to the integrated revenue change:

$$MVPF_{0\to1} \;=\; \frac{\displaystyle\int_0^1 \tau_c U_{\mathcal{T}}(\theta)\, d\theta}{\displaystyle\int_0^1 \Big[\tau_c U_{\mathcal{T}}(\theta) + \tau_c\big(1 - \theta - \tau_I\big)\tfrac{dR_{\mathcal{T}-1}}{d\theta}\Big]\, d\theta}.$$

[59] If the older generation's MPC out of terminal-period wealth is positive, our assumption $dc \approx 0$ overstates the MVPF: a positive MPC of m would add $m\tau_I \cdot dR/d\theta$ to the denominator of the MVPF. Since our central conclusion is that the MVPF is small, $dc \approx 0$ is the conservative choice.

Assuming the realization response arrives uniformly along the path ($\frac{dR_{\mathcal{T}-1}}{d\theta} = \Delta R_{\mathcal{T}-1}$ for all θ), $U_{\mathcal{T}}(\theta)$ is linear in θ, both integrands are linear, and the trapezoid rule is exact:

$$\int_0^1 \tau_c U_{\mathcal{T}}(\theta)\, d\theta = \tau_c \bar{U}, \qquad \int_0^1 \tau_c(1 - \theta - \tau_I)\Delta R_{\mathcal{T}-1}\, d\theta = \tau_c\Big(\tfrac{1}{2} - \tau_I\Big)\Delta R_{\mathcal{T}-1},$$

with $\bar{U} \equiv \frac{1}{2}(U_{\mathcal{T}}(0) + U_{\mathcal{T}}(1))$; τ_c cancels, yielding Equation 8. Without the linearity assumption, the endpoint marginal formulas bound the possible attributions of the observed response: $MVPF(0) = \frac{U_{\mathcal{T}}(0)}{U_{\mathcal{T}}(0)+(1-\tau_I)\Delta R_{\mathcal{T}-1}}$ attributes the entire response to the first increment away from step-up, and $MVPF(1) = \frac{U_{\mathcal{T}}(1)}{U_{\mathcal{T}}(1)-\tau_I \Delta R_{\mathcal{T}-1}}$ (which exceeds one when $U_{\mathcal{T}}(1) > \tau_I \Delta R_{\mathcal{T}-1}$) to the last increment before full carryover. The similarity of the short-run and medium-run responses supports the uniform-path assumption over either extreme.

C.1.2 Empirical Implementation

We implement Equation 8 with steady-state annual flows, all in 2018 dollars: $\Delta R_{\mathcal{T}-1}$ is the additional taxable realizations per year by living owners, and $U_{\mathcal{T}}(\theta)$ is the annual flow of taxable capital gains in inherited stock under regime θ. We measure both $U_{\mathcal{T}}$ endpoints directly.

Ingredient construction.

- $\Delta R_{\mathcal{T}-1}$: the age-decile ATTs from our primary age-based design (Section 5), estimated on the medium run (2009–2013, transition years 2005–2008 excluded) and multiplied by the population at each age. We use the medium-run window because it excludes the transition dynamics and matches the era of the $U_{\mathcal{T}}(1)$ measurement.

- $U_{\mathcal{T}}(\theta)$: positive gains only, because Norway's carryover was asymmetric—terminal losses were not carried over and are therefore unexposed to the reform. The measure includes *inter vivos* gifts as well as bequests. Transfers between spouses are excluded after 2006 (they were exempt from the inheritance tax and are recorded under a separate fiscal-continuity code); spousal transfers defer rather than exercise the intergenerational margin that the model prices. The baseline window is 2003–2004, because the transaction register opens at the end of 2002 and the 2002 transfer flows are mechanically incomplete; the post-reform window is 2009–2013.

- Growth adjustment: $U_{\mathcal{T}}(0)$ scales the 2003–2004 gains flow by the growth of inherited stock *values* between the two windows. This index shares its volume base with the gains numerator, and transfer values are to first order invariant to basis rules—unlike, for example, old-age equity wealth (endogenous to the reform through the portfolio-drawdown margin) or the inherited-gains flow itself (the treated outcome).

- $\tau_I = 0.10$, the top inheritance tax rate for close relatives in force during the 2009–2013 window; the pre-2009 top rate of 0.20 is reported as a variant. The effective marginal rate on inherited gains is below the top rate, and a lower τ_I lowers the MVPF, so the top statutory rate is the choice most favorable to step-up.

Remark: the static model simplification bounds the MVPF. Applying the terminal-period coefficient to responses occurring years before death treats each induced dollar as reducing eventually-inherited gains one-for-one, with the heir's forgone tax collected at face value. At face value, displacement can in fact exceed one-for-one: a dollar realized and reinvested L years before death removes the dollar itself through the basis reset, and the capital gains tax paid, τ_c, stops compounding, removing a further $\tau_c(\bar{g}-1)$ of terminal gains, where $\bar{g}$ is the gross appreciation multiple over the remaining horizon. This excess does not overturn the bound, because it is priced inconsistently if collected at face value: discounting the heir-side flows at the asset's own return (factor $\delta = 1/\bar{g}$), the effective offset per realized dollar is

$$\delta\big(1+\tau_c(\bar{g}-1)\big) \;=\; \tau_c + \frac{1-\tau_c}{\bar{g}} \;\leq\; 1,$$

with equality only at $\bar{g}=1$. The estate-shrinkage term behaves identically: the tax paid compounds to $\tau_c\bar{g}$ of missing estate at death, taxed at τ_I and discounted by $1/\bar{g}$, returning exactly the $-\tau_I$ coefficient of Equation 7. The terminal-period conventions (immediate collection, one-for-one displacement) are therefore the joint maximum of the offset under any internally consistent discounting at a rate weakly above the asset's expected return; every deviation raises FE and lowers the MVPF.

Replication inputs. Table A6 collects every input to the MVPF calculation, its units, and its source; the calculation is reproducible from this table alone.

Direction of the simplifications. Each assumption in the derivation errs in step-up's favor:

Table A6: Inputs to the MVPF Calculation

Input	Value	Definition, units, and construction
$\Delta R_{\mathcal{T}-1}$	\$447m/yr	Reform-induced taxable realizations; dollars of *gains* (not tax revenue), annual flow, 2018 USD. Listed + unlisted stock, net realized gains. Population: all individuals aged 50+ (1.76m). Medium-run (2009–13) age-decile ATTs (Figure 4a) $\times$ population at each age.
$U_{\mathcal{T}}(0)$	\$654m/yr	Counterfactual step-up inherited-gains flow; gain dollars, annual, 2018 USD. Positive gains in inherited/gifted stock, transfers to the next generation (spouses excluded). Base: 2003–04 observed flow (\$418m), scaled by growth of inherited stock *values*, 2009–13 vs. 2003–04 ($G = 1.56$).
$U_{\mathcal{T}}(1)$	\$722m/yr	Observed carryover inherited-gains flow; same concept, measured directly, 2009–13 average. No adjustment.
τ_c	0.28	Statutory flat capital gains rate, 1992–2013. Cancels from the MVPF when all flows are in gain dollars.
τ_I	0.10	Top inheritance tax rate, close relatives, 2009–13 (Table A2); pre-2009 rate 0.20 as variant.
q	1	PV fraction of heir-side capital gains tax collected; 1 = immediate realization (upper-bound convention). Variants 0.75, 0.5.
κ	0	Inheritance-tax credit per dollar of carried-over gain; Norway granted $0.2\tau_I$ (Section 3), abstracted from here (its inclusion lowers the MVPF to 0.77).
$FE_{0\to1}$	0.26	$(\frac{1}{2} - \tau_I)\,\Delta R_{\mathcal{T}-1}\,/\,\bar{U}$, $\bar{U} = \frac{1}{2}(U_{\mathcal{T}}(0) + U_{\mathcal{T}}(1)) =$ \$688m.
$MVPF_{0\to1}$	0.79	$1/(1 + FE_{0\to1})$; an upper bound (see above).

Note: $U_{\mathcal{T}}$ is gross of embedded losses while $\Delta R_{\mathcal{T}-1}$ is net of realized losses; this combination overstates the mechanical base and hence the MVPF, consistent with the upper-bound reading.

heirs are assumed to realize immediately (deferral shrinks the present value of the tax expenditure and of the heir-side revenue offset); induced realizations are credited with displacing heir realizations in full; the older generation's consumption response is set to zero; the acceleration of revenue by decades is not discounted; $U_{\mathcal{T}}$ is gross of losses while $\Delta R_{\mathcal{T}-1}$ is net; and the control group's own exposure to carryover attenuates the estimated response. The MVPF we report is therefore an upper bound, which is the basis for the main text's emphasis on the conclusion $MVPF < 1$ rather than on the point estimate.

D Data Appendix

Our paper uses transactions data to build new measures of unrealized and inherited capital gain at the individual-asset level. We detail and present several validation checks on components of these novel measures below.

D.1 Detail on Sample Selection

We exclude all individual-security ledgers with transactions indicating the borrowing or lending of shares (e.g., for short selling). Affected positions are rare: 0.41 percent of person–security combinations contain at least one borrowing or lending transaction, and 0.46 percent of individuals in the stock transactions data hold at least one such position.[60] They are, however, unusually transaction-dense, averaging roughly fifteen times as many records as an unaffected position, so the exclusion removes 5.8 percent of transaction records. That density is itself evidence that these positions represent short-term trading rather than accumulated appreciation, and are therefore unlikely to contain large unrealized capital gains. Furthermore, our alternative results using an average basis calculation (not included here for length) include these records and do not show any significant difference in any of our results.

[60]Both shares are computed on the ledger sample of personal shareholders entering our first-in-first-out matching.

D.2 Valuing Unlisted Firms in the Norwegian Data

Our preferred estimate of unlisted firm value largely follows the procedure of Smith et al. (2023) to compute a liquidity-adjusted, equal-weighted average of capitalized pro rata sales, assets, and EBITDA. Specifically, we apply a 10% liquidity discount to an equal-weighted average of three measures of the firm's capitalized value using (1) sales (2) assets and (3) EBITDA, respectively.

To capitalize each factor, we define valuation multiples for each NACE three-digit industry using Worldscope data on listed firms in the European Union, European Economic Area, and Switzerland.[61] We regress a ratio of the listed firm's enterprise value relative to the factor ($X \in \{sale, asset, ebitda\}$) on year-specific fixed effects for the firm's industry and country. We then employ the fitted values for Norway as our multiple for the given industry in the given year.

We assign unlisted firms in industries with insufficient data (less than 5 listed firms in the Worldscope data that year) a multiplier computed at a higher level in the industry classification (e.g., NACE 2, or the overall market average). We winsorize outlier multipliers at the 97.5th percentile. For firms that own other firms, we capitalize the firm value at the lowest level possible, and then sum over those valuations to value the parent firm. Since liquid financial assets are valued in the firm's book at market value, we value unlisted firms with substantial financial assets and little inherent value (holding companies) simply using the book values of assets minus liabilities. Finally, the equal-weighted capitalization approach can undervalue asset-heavy firms (e.g., holding companies and real estate firms) where income and sales multipliers are inappropriate. We therefore set a floor for unlisted firm value at the (book-based) tax valuation.

D.3 Robustness to Alternative Basis Calculation

Norway's tax code specifies that capital gains are computed on a first-in, first-out (FIFO) basis, and our primary empirical estimates accordingly employ a FIFO cost basis constructed by matching each asset disposal to its original acquisition in the transactions data. However,

[61] Limiting ourselves to only Norwegian listed firms does not allow sufficient industry variation to estimate multipliers for all the industries of unlisted firms.

implementing FIFO at the transaction level requires normalizing nominal share counts across stock splits and consolidations so that acquisitions and dispositions can be matched on either side of such changes. This normalization step introduces a potential source of measurement error: if the imputed split or consolidation multiplier is incorrect for a given firm-date, the matching algorithm may fail to link the appropriate lots, and the resulting basis calculation will inherit the error. The computational burden of the FIFO matching procedure is also substantial. In the model of Section 4, tractability demands that we instead adopt an average basis assumption.

As a robustness check, we therefore also compute unrealized capital gains under an average cost basis assumption, in which the cost basis of disposed shares equals the weighted average acquisition cost of all shares currently held. The average basis approach is both simpler to implement and computationally faster, as it requires only a running weighted average rather than lot-level matching across transactions. It also avoids the issue of stock split and consolidation adjustments. As Figure A8 demonstrates, aggregate unrealized capital gains are nearly perfectly aligned across the two methods, suggesting that the choice of cost basis rule does not materially affect our main estimates.[62]

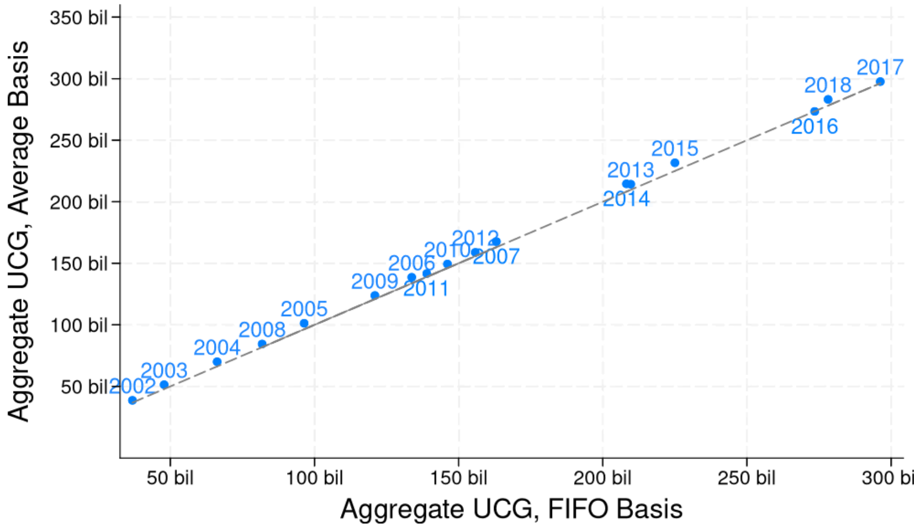
D.4 Validating End-of-Year Stock Holdings

From 2004, the Norwegian Shareholder Registry documents the ownership shares by Norwegian shareholders of AS and ASA (limited liability) companies. We can therefore compare the ownership amounts documented in the Shareholder Registry data with our transaction-based calculation of individual shareholders' end-of-year holdings in these companies.

The two measures of ownership agree very closely: for AS and ASA companies subject to Shareholder Registry reporting, we are able to match 97% of Norwegian shareholders in our data into the Shareholder registry, and 99.6% of all matched person-year observations agree perfectly between the two data sources. That is, our transaction-based calculation of a given individual's number of shares owned at the end of the year in a given company usually agrees perfectly with the number of shares that the Shareholder Registry says that

[62] Aggregate UCG under the average basis calculation is a little larger that under FIFO, as would be expected if the "first in" positions contain higher than average gains

Figure A8: Aggregate Unrealized Capital Gains (UCG) Comparison, FIFO vs. Avg. Basis



Note: This figure compares our series for the aggregate amount of unrealized capital gains in stock calculated with our baseline first-in-first-out (FIFO) basis assumption (x-axis, in billions of 2018 USD) to the same series calculated with an average basis assumption (y-axis) from 2002 to 2018. As a point of reference, the dashed line represents perfect agreement (y=x).

individual owns in that company. Residual disagreement (illustrated in Appendix Figure A9 Panel A) is probably due to issues with merging the two datasets, or limitations in our procedures to account for mergers and other company transformations.

D.5 Comparing Realized Capital Gains to Tax Returns

Our measures of unrealized and inherited capital gains are novel, and thus hard to compare to existing estimates. In contrast, taxable realized capital gain is a component of taxable income, and is documented each year in every individual's tax return. Validating our calculation for realized capital gain allows us to conclude that our measure of basis is sensible, and that the transactions data reflects realizations in the tax return.

However, we are not able to make an exact comparison between two measures of realized capital gain in stock. The individual tax returns report total taxable capital gains and losses and real estate-specific taxable gains and losses. Below, we compare net taxable realized capital gains from assets other than real estate to our measure of realized capital gains from directly-held stock (listed and unlisted), net of the rate-of-return deduction. These measures may differ if the individual realized capital gains or losses in asset classes other than real estate or directly-held stock (e.g., from mutual funds). It may also differ (although probably only slightly) because individuals can deduct transaction costs (such as broker fees) from their taxable gain, and we do not observe these costs in our data.

Despite these known differences, the two measures of realized gain align well. 97% of all individuals in the tax data report net realized capital gain from non-real estate that is within rounding error (2000 NOK) from our calculation in the stock transactions data. Appendix Figure A9 Panel B illustrates the comparison.

D.6 Comparing Aggregate Values to Andresen (2022)

Andresen (2022) presents aggregate measures of unrealized capital gains for unlisted companies in Norway from 2006-2016. This Statistics Norway report is the only other work we are aware of that makes such a calculation. Andresen calculates unrealized capital gains as equal to the value of the unlisted stock minus basis and rate-of-return deduction. He predicts the market value of unlisted stock by multiplying book value times a book-to-price multiplier calculated from listed Norwegian companies. He has a direct measure of basis and rate-of-return deductions from the Norwegian Tax Authority (data we ourselves do not have access to), but acknowledges large problems and unlikely outliers in this measure.[63] Despite these data issues and the different procedure, his aggregate amounts of unrealized capital gains are broadly similar to our own estimates for unlisted firms (see Appendix Figure A10).

E Benchmarking Our Realization Response Against Rate Elasticities

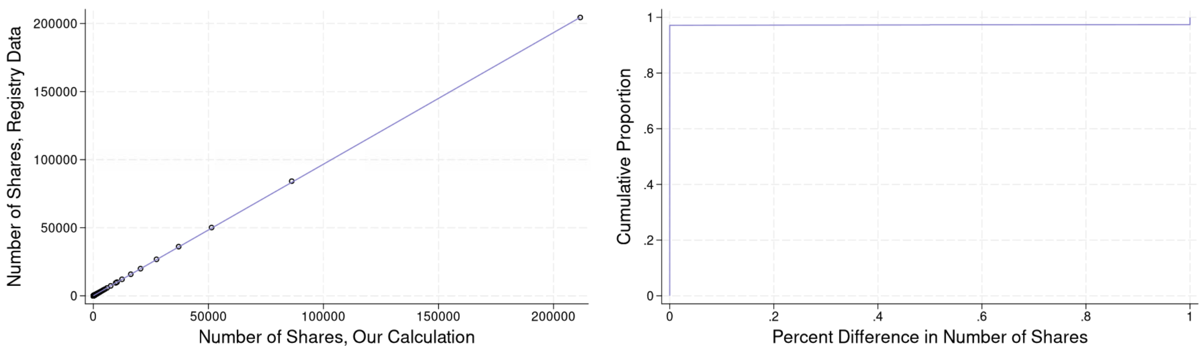
The removal of step-up raised taxable realizations without changing the statutory capital gains tax rate. To gauge the magnitude of this response, we ask: how large a cut in the

[63] To quote from the report regarding the issue with the Norwegian Tax Authority's data on basis (not used in our own analysis): "The data on basis unfortunately has several weaknesses. In many cases, the basis values will be too large [...] Somewhat surprisingly, there is also a significant number of negative basis values in this data set, and for the years 2006-2009 this is so common that the sum of all basis values for unlisted shares is negative [...] it is unlikely that these negative basis values are real." (Andresen, 2022) (authors' own translation)

Figure A9: Validation Exercises

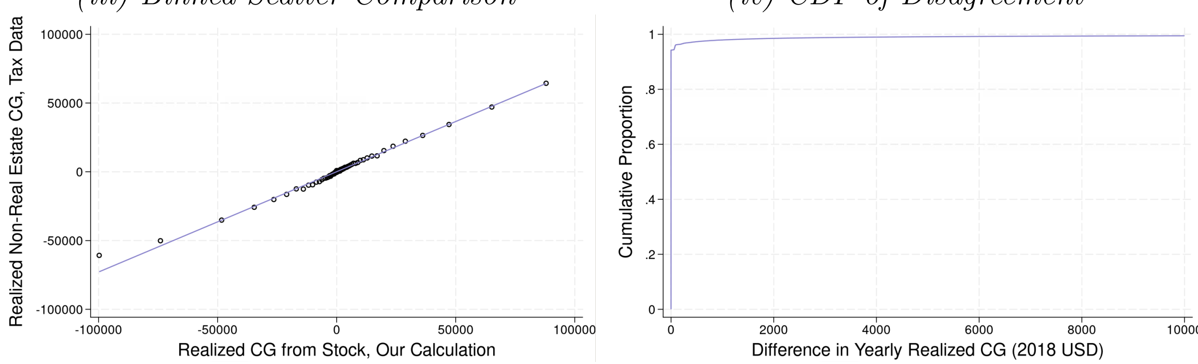
Panel A: End-of-Year Share Comparison

(i) Binned Scatter Comparison *(ii) CDF of Disagreement*



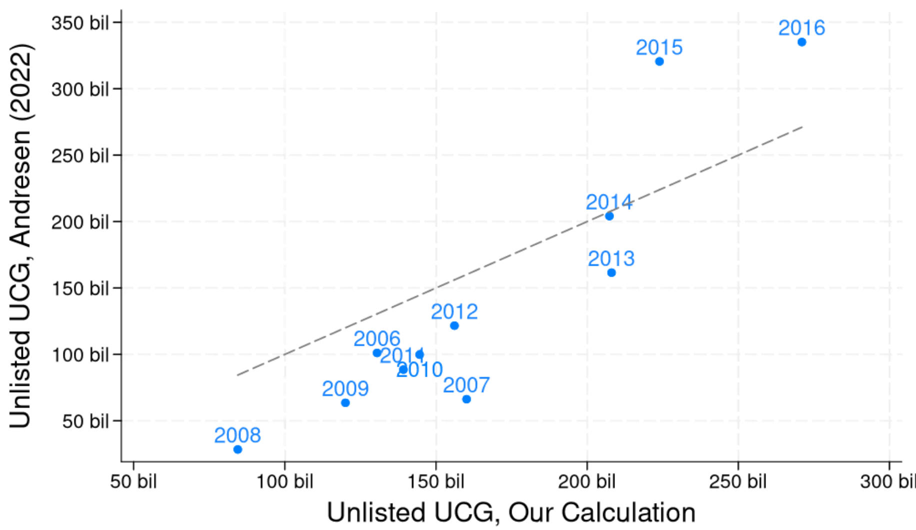
Panel B: Realized Capital Gains Comparison

(iii) Binned Scatter Comparison *(iv) CDF of Disagreement*



Note: Panel A compares our calculation of end-of-year shareholdings (at the individual-company-shareclass-year level) to stock ownership reported in the Shareholder Registry. This panel includes every shareholding of AS and ASA companies in our data from 2004-2018, whether or not we are able to match that shareholding into the Registry. 97% of all shareholdings match and agree exactly between the two data sources. Failures to find the shareholding (i.e., the year-person-company-shareclass record) in the Registry are coded as having zero Registry shares, and these failures to match account for the residual disagreement illustrated in Subfigure (ii). Panel B compares our calculation of individuals' yearly net realized capital gains in stock to non-real estate net realized capital gains reported in individuals' tax returns. This panel includes every person with an individual tax return from 2003-2004 and 2006-2018 (2005 is excluded due to our inability to separate out real estate realizations in the tax return data that year). 97% of all yearly realizations agree within rounding error. As depicted in Subfigure (iii), larger disagreements are associated with higher values in the tax returns than our own transaction-based calculation, consistent with the fact that the tax return-based measure includes realizations from non-real estate assets other than directly held stock.

Figure A10: Aggregate Unrealized Capital Gains (UCG) Comparison with Andresen (2022)



Note: This figure compares our series for the aggregate amount of unrealized capital gains in unlisted companies (x-axis, in billions of 2018 USD) to the same series from Andresen (2022) (y-axis) from 2006 to 2016. As a point of reference, the dashed line represents perfect agreement (y=x). The series from Andresen (2022) is taken from Figure 2(a) of that report, where it is reported (in 2021 kroner) as "skattepliktige latente eierinntekter" (taxable latent owner income), using the method of valuing unlisted stock with book-to-price multipliers. Values are then converted into 2018 dollars for comparison with our estimates. It is worth noting that Andresen (2022) expresses substantial reservations about the quality of his data on basis. Given the data and methodological differences between our approaches (detailed in Appendix Section D.6), we consider the depicted level of agreement to be reasonable.

capital gains tax rate would have been required to generate a realization response of the same size, given realization elasticities estimated in the quasi-experimental literature? This appendix formalizes that conversion.

Setup

Let $R(\tau)$ denote aggregate taxable realizations as a function of the capital gains tax rate τ, and let $\tau_0 = 0.28$ be the Norwegian capital gains tax rate throughout our baseline period. Our reduced-form estimates imply that the removal of step-up increased taxable realizations by $g = 17\%$ relative to counterfactual.[64] The rate τ^{eq} that would deliver an equivalent response solves $R(\tau^{\text{eq}})/R(\tau_0) = 1 + g$, which requires specifying the realization function $R(\tau)$ over the interval between τ^{eq} and τ_0. Rather than impose a single functional form, we take two recent elasticity estimates from the literature together with the (different) functional forms used to create each estimate.

Our first benchmark, Agersnap and Zidar (2021), estimates a constant elasticity of realizations with respect to the net-of-tax rate:

$$R(\tau) \;=\; R(\tau_0)\left(\frac{1-\tau}{1-\tau_0}\right)^{\varepsilon}, \tag{10}$$

under which the equivalent rate is

$$\tau^{\text{eq}} \;=\; 1-(1-\tau_0)\,(1+g)^{1/\varepsilon}. \tag{11}$$

Our second benchmark, Dowd and McClelland (2024), estimates a semi-logarithmic form:

$$\ln R(\tau) \;=\; \ln R(\tau_0) \;-\; b\,(\tau-\tau_0), \qquad b \;>\; 0, \tag{12}$$

under which the equivalent rate is

$$\tau^{\text{eq}} \;=\; \tau_0 \;-\; \frac{\ln(1+g)}{b}. \tag{13}$$

[64] Because the estimated response is stable across our post-period, with no decay through 2018 (Section 5.3), we interpret g as the permanent annual response and accordingly compare to long-run elasticity estimates.

To keep magnitudes comparable across the two benchmarks despite the different forms, Table A7 also reports each calibration's implied elasticity with respect to the tax rate evaluated at the common Norwegian rate $\tau_0 = 0.28$.

Choice of Elasticities

For Agersnap and Zidar (2021) we use their preferred log-log estimates directly. Their long-run (post-reform years 6–10) estimates across three specifications are $\varepsilon = 1.47$ (baseline), 1.18 (controlling for other state tax changes), and 0.99 (identifying only from reforms larger than one percentage point). At $\tau_0 = 0.28$, these imply local tax-rate elasticities of 0.39 to 0.57.

We also use the long-horizon (four years of lags) specification of Dowd and McClelland (2024) (an update of Dowd et al. (2015)). Because their model is semi-log, their permanent elasticity of -0.59 is evaluated at their sample's rates and must be restated at the Norwegian rate: dividing by the mean marginal tax rate of 17.4 percent among realizers in the same panel recovers the coefficient $b = 0.59/0.174 = 3.39$, implying a local elasticity of 0.95 at $\tau_0 = 0.28$.[65]

It is worth noting that Gravelle (1991, 2021) argue that permanent elasticities above roughly 0.34 are infeasible, which would rule out the larger calibrations here. Adopting that bound would only strengthen our conclusion: a smaller elasticity implies a larger rate cut is needed to match our estimated response, and hence a larger revenue loss from the equivalent rate-cut policy.

Results

Table A7 reports τ^{eq} from Equations (11) and (13). Matching the aggregate realization response of 17 percent would have required cutting the Norwegian capital gains tax rate from 28 percent to 23.4 percent under the Dowd–McClelland benchmark, and to between

[65] Dowd and McClelland (2024) do not report the exact rate at which their elasticity is evaluated; we use the weighted mean marginal rate among realizers reported for the same 1999–2008 panel in Dowd et al. (2015). Note that restating a 28-percent-rate elasticity this way assumes the linear dose-response of Equation (12) holds between the sample mean and 28 percent; combined federal-plus-state rates of 28 percent are within the support of their individual-level data.

19.9 and 15.6 percent under the Agersnap–Zidar estimates—that is, eliminating between a sixth and nearly half of the rate.

These comparisons also imply an immediate contrast in revenue expectations. Since $R(\tau^{\text{eq}})/R(\tau_0) = 1 + g$ by construction, capital gains tax revenue at the equivalent rate, relative to revenue at τ_0, is

$$\frac{\tau^{\text{eq}}\, R(\tau^{\text{eq}})}{\tau_0\, R(\tau_0)} \;=\; \frac{\tau^{\text{eq}}}{\tau_0}\,(1+g) \tag{14}$$

under either functional form. Whether this ratio falls below one depends on the revenue-maximizing rate, which both studies' baseline estimates put above the Norwegian rate of 28 percent. Under either paper's implied Laffer curve, the equivalent rate cut would therefore have reduced capital gains tax revenue (by 2 to 35 percent across the calibrations in Table A7), whereas the removal of step-up generated the same realization response while raising revenue by 17 percent. The economic content of this contrast does not depend on the exact revenue figures: a rate cut purchases its realization response by lowering the rate on all inframarginal realizations, while removing step-up raises realizations by eliminating a subsidy to deferral – the targeting difference we return to in Section 6.3.

Table A7: Rate Cut Required to Match the Realization Response to the Removal of Step-Up

Calibration	Functional form	Elasticity at $\tau_0 = 0.28$	Equivalent rate τ^{eq}	Implied cut (pp.)	Revenue change
A–Z, baseline	log-log ($\varepsilon = 1.47$)	0.57	19.9%	8.1	−17%
A–Z, other-tax controls	log-log ($\varepsilon = 1.18$)	0.46	17.7%	10.2	−26%
A–Z, large reforms only	log-log ($\varepsilon = 0.99$)	0.39	15.6%	12.4	−35%
Dowd–McClelland (2024)	semi-log ($b = 3.39$)	0.95	23.4%	4.6	−2%

Note: This table reports the capital gains tax rate τ^{eq} that would generate a realization response equal to our estimated response to the removal of step-up. Each calibration is evaluated under the functional form in which it was estimated: for Agersnap and Zidar (2021) ("A–Z"), the constant net-of-tax-rate elasticities ε of Equation (10), taken directly from the post-reform years 6–10 rows of their Table 2 (baseline, controls for other state tax changes, and identification from reforms exceeding one percentage point, respectively)—the estimates underlying their revenue-maximizing rates of 38–47 percent—with τ^{eq} from Equation (11); for Dowd and McClelland (2024), the semi-log coefficient b of Equation (12) implied by their four-lag permanent elasticity of −0.59 evaluated at the mean marginal rate of 17.4 percent among realizers in the same panel (Dowd et al., 2015), $b = 0.59/0.174 = 3.39$, with τ^{eq} from Equation (13). The elasticity column reports the implied local elasticity of realizations with respect to the tax rate at the Norwegian baseline rate $\tau_0 = 0.28$: $\varepsilon\,\tau_0/(1-\tau_0)$ for the log-log rows and $b\,\tau_0$ for the semi-log row. Revenue changes are computed from Equation (14) and refer to capital gains tax revenue only.

Caveats

Three caveats apply. First, these elasticities are identified from U.S. variation, predominantly state-level tax changes (Agersnap and Zidar, 2021) and individual federal-plus-state rate variation (Dowd and McClelland, 2024); portability to Norway is imperfect. Second, the required cuts lie below the support of the identifying variation in these studies, so τ^{eq} should be read as an order-of-magnitude benchmark rather than a point prediction. We mitigate this concern by evaluating each estimate under its own functional form and by showing that the functional-form choice does not overturn the conclusions. Third, the comparison holds fixed the composition of realizations, whereas a rate cut and the removal of step-up target different margins: the former lowers the price of all realizations, the latter only the return to deferring gains until death. This difference in targeting is exactly why the two policies, despite equal realization responses, have opposite revenue effects. It is also why the elasticities from a tax regime with stepped-up basis are the conceptually appropriate ones for this exercise: the pre-2005 Norwegian baseline from which the hypothetical rate cut would depart.

F Background on Norway

F.1 Taxation in Norway, 1992-2018

Norwegians pay taxes on their worldwide labor income, capital income, and wealth. Labor income is subject to progressive taxation with a top marginal rate of 53.2–54.3% (including social security contributions) during our period of study. Net wealth in excess of an exemption amount faces a yearly tax of 0.85–1.30% during our period.[66]

[66] The wealth tax exemption has ranged from $19,555 to $162,800 (2018 USD) during our period of study. In general, exemptions have grown more generous and wealth tax rates have fallen over our period of study. See Thoresen et al. (2022) for a detailed description of Norway's wealth tax. Interestingly, Norway collects a smaller share of total tax revenue from property and wealth taxation than is typical among OECD countries (OECD, 2018).

F.1.1 Capital Gains and Inheritance Taxation

Norway has a flat capital gains tax on realized capital gains. From 1992 to 2013, all (net) realized capital gains were taxed at a rate of 28%.[67] Since 2014, there have been several rate changes, including the introduction of separate rates for the realization of capital gains from stock and non-stock assets. These rate changes are depicted in Figure A1. Throughout our study period, the capital gains tax has had a flat rate structure, so that capital gains tax rates do not depend on individual income.

Prior to its abolition in 2014, Norway's integrated gift and inheritance transfer tax taxed the receipt of gifted or inherited assets through a rate structure that depended on the cumulative amount of the gifts/inheritance received, and the relationship between the giver and the recipient. Other than a small yearly gift exemption, no distinction in tax rates was made between *inter vivos* gifts and bequests at the time of death. Appendix Table A2 provides details on the rates and brackets over time. Figure A1 depicts the top rate for gifts and inheritances between close relatives: a rate that fell from 20% to 10% in 2009.[68] In 2014, the gift and inheritance tax was abolished, so all rates for subsequent years are set to 0%.

F.1.2 Stepped-Up Basis Policy

During our period of study, Norway moved from a system of stepped-up basis for inherited capital gains to a system of carryover basis. This change was accomplished in two steps. In 2006, carryover basis was introduced for appreciated gifted or inherited stock (including equity-based mutual funds). Recipients of gifted or inherited stock from 2006-2013 continued to be subject to inheritance tax on value of the stock they received, but now were also subject, if and when they sold the stock, to capital gains tax on the amount of appreciation during the original owner's holding period. Losses, however, were not permitted to be carried over in the 2006 reform – so stocks with losses continued to be subject to a "step-down" in basis. In 2014, concurrent with the abolition of the gift and inheritance tax, carryover basis was

[67] This rate is also applied to other forms of capital income, such as interest income. Capital losses can be used to fully offset capital gains and ordinary income (but not social security contributions or top marginal tax rates).

[68] The definition of close relatives includes parents and children of the giver. Transfers between spouses were not taxed. Transfers between individuals who are not close relatives were subject to a higher top rate in every year: 30% before 2009, and 15% until 2014.

introduced for other assets, including stocks with losses. Therefore, since 2014, recipients of gifted and inherited assets do not pay inheritance tax at the time of the receipt, but are subject to capital gains tax on appreciation from the original owner's holding period when they sell the inherited asset.

Certain types of real estate are exempt from capital gains taxation in Norway if the owner's use of the property and length of ownership meets certain requirements. Most prominently, residential property is not subject to capital gains tax if the seller has owned and lived in the property as his primary residence for 12 of the last 24 months. Vacation homes can be similarly exempt if the seller has owned the property for more than five years and used it as his own vacation home for at least five of the last eight years. Buildings on a farm may be exempt if the farm has been owned for over ten years. In these cases in which the original owner could have sold the property without capital gains tax, the heir's basis continues to be stepped up to the assumed sales value at the time of acquisition, even after 2014.

F.1.3 Legislative History & Anticipation of Step-Up's Removal

Norway's policy of stepped-up basis for gifted and inherited assets was first established by Norwegian Supreme Court rulings in the 1920s. Various government committees sporadically proposed moving to a system of carryover basis (proposals were made in 1975, 1981 (for gifts only), 1989 and 2000), but no changes were enacted until 2006 (Zimmer, 2014).

In 2005, it was announced that step-up would be removed for stock (including unlisted stock and equity-based mutual funds).[69] The 2006 removal of stepped-up basis for inherited stock was passed as an afterthought to a wider reform of capital taxation. The 2006 tax reform changed the exemption rules for dividend and capital gains taxes in a way that made

[69] The removal of step-up for stock was formally introduced on December 10, 2004 (Stortinget, 2004). However, this change did not receive immediate media attention, so we take 2005 to be our announcement year. A search through the Norwegian media library (Nasjonalbiblioteket) for the terms "kontinuitetsprinsippet" and "diskontinuitet" (which are the Norwegian terms corresponding to carryover and stepped-up basis, respectively) shows that the first mention of the impending change appeared in the newspaper *Finansavisen* on January 8, 2005, under the headline "One Year Left" (Amundsen, 2005). A second mention appeared in *Levanger-Avisa* on February 19, 2005 (Vigdal, 2005). Both articles featured lawyers advocating that Norwegians transfer stock (especially stock of family businesses) to the younger generation before the new carryover basis regime came into effect. Media coverage of the upcoming change grew more intense during the spring and summer of 2005.

these exemptions more dependent on owners' cost basis (see Section F.1.4). Because of concern that owners might transfer shares as gifts to their relatives to get the increased dividend exemptions that would accompany stepped-up basis, subsequent legislation was passed that abolished stepped-up basis for shares starting in 2006. Although the main components of the 2006 tax reform were formally announced in March 2004 and approved in June of that year, the removal of step-up was not part of the June legislation and was not definitively decided until later in the year (Norwegian Ministry of Finance, 2004).

The 2014 removal of step-up for other assets was closely tied to the abolition of the inheritance tax. Right-wing parties campaigned on abolishing the inheritance tax during the 2013 election, which they won in September 2013. Therefore from September 2013, it was reasonably certain that the inheritance tax would be repealed, although the exact timing of the repeal initially remained uncertain. The rules accompanying that repeal, which included the removal of step-up for non-stock assets, were announced as part of the government's yearly budget in October 2013.

F.1.4 Other Aspects of the 2006 Reform

As mentioned above, the removal of step-up for stock came about due to a larger overhaul of exemptions for capital gains and dividend taxation. The overall reform was primarily motivated by a need to prevent private business owners from relabeling labor income as dividends, which had become endemic to the old system. Under the old system (called "RISK"), both capital gains and dividends were given generous exemptions that corresponded to the amount of retained earnings held in the underlying firm. The goal of this exemption was to avoid double-taxation on corporate earnings which would have already been subject to the corporate tax. Furthermore, dividend distributions were subject to a yearly limit that corresponded to the amount of retained earnings in the firm at the beginning of the year. Because the exemption amount and dividend cap were thus linked, the effect was that almost no dividends were taxed prior to 2006. From 2006, the same dividend cap remained in place, but the exemption calculations for dividends and capital gains were made less generous going forward. Henceforth, investors would accumulate exemptions based on how their costs basis would grow with the normal rate of return.

Previous work has documented that the 2006 reform led to a large fall in dividend distributions (Alstadsæter and Fjærli, 2009; Alstadsæter et al., 2014, 2019). However, we do not think that this aspect of the reform is likely to be linked to the increase in capital gains realizations we document later in this paper (Section 5). First of all, our results in Section 5 show a response starting in the announcement year of 2005, before the exemption calculations changed.[70] Secondly, the old exemption amounts remained usable for realizations after 2006 – only the accumulation of future exemption amounts changed.[71] Therefore, there was no immediate change to the tax calculation of realizing capital gains in 2005 vs. early 2006 – the same exemption amounts would be available both times. Finally, we do not think there is any incentive for investors to shift from dividend distributions to capital gains realizations. Both before and after the reform, dividends and capital gains were subject to the same tax rate and granted the same exemption amounts. Investors should therefore be "neutral" between the two methods of extracting money from the firm both before and after the reform (Sørensen, 2005).[72] We therefore do not expect any of our documented increase in capital gains realizations (Section 5) to be a shift from dividends to capital gains realizations.

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[70]It is important to recall that our realization results look at the outcome of *taxable* realized capital gains, and so do not include any (tax-free) company transformations occurring under Transition Rule E.

[71]See The Norwegian Tax Administration (n.d.): "For shares acquired before 1 January 2006, the input value is adjusted for accumulated RISK."

[72]See Bjerksund and Schjelderup (2021a,b) for discussions of how exemptions and loss limitations affect the capital gains realization decision in Norway.

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